Performance Benchmarking of Deep Learning Models on BdSLW-11 for Bangladeshi Sign Language Word Recognition

A submitted thesis by

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CERTIFICATION OF ORIGINALITY

I, Rafiul Islam, a student of the Department of Computer Science & Engineering at Pundra University of Science & Technology, Bogura-5800, Bangladesh, hereby declare that this thesis is the result of my own independent work. It has not been submitted previously, either in whole or in part, for the award of any degree, diploma, or certificate at this or any other academic institution.

Any sources of information or assistance used during the preparation of this thesis—whether from books, research articles, websites, or individuals—have been properly cited and acknowledged. While I have received minor support in terms of language refinement and formatting guidance, all core concepts, analysis, and research contributions presented in this work are solely my own.

This thesis reflects my personal understanding, effort, and academic integrity.

……

Rafiul Islam

CERTIFICATION OF APPROVAL

This is to say that the thesis written by Rafiul Islam, named:“Performance Benchmarking of Deep Learning Models on BdSLW-11 for Bangladeshi Sign Language Word Recognition”has been checked and found good. It follows all the university rules. The work is original and done by him. The quality of the work is okay and not copied from anyone.
This thesis was done for getting the degree from the Department of Computer Science & Engineering at Pundra University of Science & Technology, Bogura-5800, Bangladesh.

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Mrittika Mahbub

Lecturer

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ABSTRACT

Globally, millions of people experience hearing impairments, including over 2.6 million individuals in Bangladesh alone. For these communities, sign language serves as a primary mode of communication. However, in the context of Bangladesh, technological advancements supporting Bangladeshi Sign Language (BdSL) remain significantly limited. Most of the existing research has focused on alphabet-level recognition using the BdSL49 dataset and single CNN-based deep learning models. These efforts, while valuable for foundational work, are insufficient for building real-world applications that require full-word level sign interpretation. Moreover, prior studies often relied on simple validation techniques and failed to report key performance indicators like F1-score, AUC (Area Under Curve), false positive and false negative rates, and training time, which are essential for assessing model robustness and practical deployment feasibility.

To overcome these limitations, this study proposes a complete deep learning framework for the recognition of full Bangla sign words using the newly introduced BdSLW-11 dataset. This dataset consists of 1,105 manually labeled RGB images representing eleven of the most frequently used words in daily BdSL communication, such as “Friend,” “Beautiful,” “Water,” and others. To train the models efficiently, we adopted the transfer learning approach on six widely used convolutional neural network (CNN) architectures: VGG16, VGG19, ResNet50, InceptionV3, MobileNetV3, and EfficientNetB3. A comprehensive data augmentation pipeline was applied during preprocessing to enhance the model’s generalization capabilities and simulate real-world visual variations.

A key methodological strength of our research is the implementation of **Stratified 10-Fold Cross Validation**, which ensures fair and balanced training-validation splits across all classes in the dataset. Unlike previous approaches that use a single static train-test split, our cross-validation strategy reduces variance and increases the reliability of the reported metrics. Performance was measured using multiple evaluation parameters, including average validation accuracy, F1-score, AUC, FPR (False Positive Rate), FNR (False Negative Rate), and training time for each model. Additionally, confusion matrices and ROC (Receiver Operating Characteristic) curves were used for visual performance analysis and better interpretability.

Among all tested models, **EfficientNetB3** stood out as the best-performing model, achieving an average validation accuracy of **97.8%**, F1-score of **0.979**, and an exceptionally high AUC of **0.999**. It also maintained the lowest false positive rate (0.0022) and false negative rate (0.0217), indicating superior reliability in both detection and classification. **InceptionV3** and **ResNet50** followed, with validation accuracies of **95.3%** and **93.9%** respectively, while **VGG16** and **MobileNetV3** performed moderately well. In contrast, **VGG19** demonstrated the lowest overall performance in terms of accuracy, F1-score, and error rates, despite having one of the highest training times.

.This research contributes three significant advancements: (1) the first multi-model benchmark using BdSLW-11 for full-word classification; (2) rigorous evaluation using 10-fold stratified validation; and (3) detailed metric reporting for reliable comparison and future development. By shifting from letter-level detection to word-level recognition, the system marks a meaningful step toward real-world BdSL integration. Future work includes optimizing the models for live video input and deploying them in mobile, educational, or assistive tools tailored for the Deaf and Hard-of-Hearing (DHH) community in Bangladesh.

Keywords*: Bangladeshi Sign Language (BdSL), BdSLW-11 Dataset, Sign Language Recognition, Deep Learning, Word-level Detection, Transfer Learning, EfficientNetB3, InceptionV3, VGG16, VGG19, ResNet50, CNN Models, K-Fold Cross Validation, Assistive Technology, Computer Vision.*

Table of Contents

FIGURES

TABLES

Introduction

1.1 Background and Introduction

Verbal communication is one of the most fundamental and effective ways for humans to express emotions, convey information, and fulfill their daily social needs. However, individuals who are Deaf or Hard-of-Hearing (DHH) do not have access to this common communication channel. To overcome this barrier, they use sign language—a structured visual language that incorporates hand gestures, facial expressions, and body postures to represent words and ideas. Unlike spoken language, sign language is purely visual and spatial, making it uniquely powerful for those with hearing impairments.

In the context of Bangladesh, the official sign language used by the DHH community is known as Bangladeshi Sign Language (BdSL). According to the World Health Organization (WHO), more than 2.6 million people in Bangladesh are affected by some form of hearing loss. Recognizing the importance of communication access for this marginalized group, the Government of Bangladesh officially recognized BdSL in 2009 as a mode of communication for DHH individuals [1], [4]. Despite this legal acknowledgment, the use of BdSL remains limited in mainstream institutions such as educational settings, healthcare facilities, public service offices, and workplaces.

This gap in accessibility results in significant social and economic exclusion of DHH individuals. Without effective communication channels, they often struggle to access basic services, participate in community life, or assert their rights. Therefore, it becomes essential to develop intelligent, technology-driven systems that can bridge this communication gap. Leveraging modern technologies such as Artificial Intelligence (AI) and Deep Learning (DL), we can build automated tools capable of translating sign language into textual or spoken formats, thereby enabling seamless interaction between DHH individuals and the rest of society [3], [5].

1.2 Importance of Sign Language Recognition

Technological advancements in computer vision and deep learning have enabled machines to understand human gestures, particularly sign languages, with increasing accuracy. Countries like the United States and the United Kingdom have already deployed real-time sign language recognition systems capable of interpreting American Sign Language (ASL) and British Sign Language (BSL) in various settings such as airports, classrooms, and customer service kiosks [5]. These tools significantly enhance the quality of life for DHH individuals by providing them access to communication without human interpreters.

However, in Bangladesh, similar assistive technologies are not yet mainstream. Most existing efforts focus on recognizing isolated Bangla alphabet signs from datasets like BdSL49 [2], [3]. While these models have shown high accuracy in recognizing individual letters, they fall short of facilitating practical communication. Real-world interaction requires the recognition of entire words and phrases, not just alphabets.

This research addresses this critical gap by focusing on the recognition of full Bangla sign words. By designing a system that can interpret words such as "Water," "Eat," or "Beautiful," we aim to offer a solution that mirrors real-life communication more closely. Such a system would have transformative impacts on how DHH individuals engage in educational, healthcare, and public service domains [4], [7].

1.3 State of the Art

Deep learning has revolutionized the field of image recognition, particularly through the use of Convolutional Neural Networks (CNNs). CNN architectures like VGG16, ResNet50, MobileNetV3, and EfficientNetB3 are renowned for their ability to extract hierarchical features from images and classify complex visual patterns with high accuracy [1], [5]. These models have been effectively applied in various international research projects involving sign language recognition.

In Bangladesh, previous research largely focused on recognizing hand gestures corresponding to Bangla alphabet letters using the BdSL49 dataset [2]. For instance, Islam et al. (2019) employed MobileNetV2 to recognize alphabet gestures with a training accuracy exceeding 98% [1]. Hoque et al. (2020) also applied a basic CNN architecture for Bangla hand sign detection, achieving similarly high accuracy but only on individual letters [2].

While these studies mark significant steps in local research, they lack real-world utility due to their limited scope. Recognizing only alphabetic signs does not suffice for meaningful conversation. Recognizing this, Chowdhury et al. (2022) introduced the BdSLW-11 dataset—a new benchmark consisting of 1,105 images spanning eleven complete Bangla sign words [4]. Each class in this dataset includes diverse images captured under varying conditions and with multiple signers, providing a robust base for training advanced models capable of word-level recognition.

1.4 Motivation of the Research

Despite the official recognition of BdSL, technological advancements in Bangladesh have yet to adequately support DHH communication needs. Most available systems focus on recognizing alphabets, which is inadequate for facilitating fluid conversation. Moreover, previous works often explored only one model architecture without comprehensive comparative analysis, making it difficult to identify the most effective solutions [1], [2], [4].

This research aims to bridge this gap by focusing on full-word recognition using the BdSLW-11 dataset. Another key motivation is the underutilization of BdSLW-11, despite its potential for real-world applications. To ensure robustness, we evaluate five well-established CNN models—VGG16, ResNet50, InceptionV3, MobileNetV3, and EfficientNetB3. We use transfer learning to fine-tune these models on BdSLW-11 and employ a rigorous evaluation strategy using Stratified K-Fold Cross Validation. This allows us to make a fair and balanced comparison of their performance [5], [6].

1.5 Research Aims and Objectives

The primary objective of this research is to build a robust, deep learning-based system capable of recognizing full Bangla sign words from static images. Such a system can dramatically improve communication for DHH individuals in educational institutions, healthcare centers, and public service environments.

To achieve this objective, we selected the BdSLW-11 dataset [4]. This dataset was cleaned and preprocessed to standardize image dimensions and normalize pixel values. Data augmentation techniques such as random rotation, flipping, and zooming were applied to artificially increase dataset size and improve model generalization.

We employed five CNN models using transfer learning: VGG16, ResNet50, InceptionV3, MobileNetV3, and EfficientNetB3. Each model was trained and validated on the dataset, and their performance was assessed using accuracy, precision, recall, F1-score, AUC, confusion matrix, and training time. Additionally, we assessed the feasibility of each model for real-time applications by considering their speed and resource requirements [5], [6].

1.6 Why We Used Stratified K-Fold

The BdSLW-11 dataset has a small and imbalanced number of images across its 11 classes. Random train-test splitting may result in unfair distributions — with some classes appearing more often in the training set and less in testing, leading to biased learning. This problem was addressed by using Stratified K-Fold Cross Validation.

Stratified K-Fold ensures that each fold maintains the same percentage of classes. For example, if one class makes up 10% of the total dataset, it remains 10% in both training and test sets of each fold. This method helps train the model more fairly and provides a better average evaluation across different data splits [4], [6].

We used 10-fold stratification, which means the model was trained and evaluated 10 times on different parts of the dataset. This also helps to avoid overfitting and gives more trusted results about how well the model works on new data.

1.7 Contributions of the Study

This study provides several important contributions to the field of BdSL recognition. First, our system can recognize full sign words — something very few studies in Bangladesh have achieved. Earlier research focused mainly on letters [1], [2], but our model can detect words like "Water", "Skin", or "Friend", which makes it more practical and helpful in real life [4].

Second, we tested and compared five CNN models — including EfficientNetB3 and MobileNetV3 — giving valuable insights into model efficiency, accuracy, and suitability for real-time use. EfficientNetB3 showed the best performance, while MobileNetV3 was fastest and lightweight for mobile use [5], [6].

Third, we used a robust method called Stratified K-Fold Cross Validation to ensure fair and balanced model testing. Many older works used random splits and didn’t report essential evaluation metrics like AUC or F1-score [1], [3]. We included these metrics, along with confusion matrices and training time.

Fourth, we helped prepare the BdSLW-11 dataset through cleaning, augmentation, and structuring. This provides a baseline for future research and helps standardize work in this area [4].

Lastly, we proposed future work using YOLOv8 for live sign recognition. While our current system works on images, we plan to build a real-time system using video input to further support the DHH community in Bangladesh [5]

CHAPTER TWO

LITERATURE REVIEW

2.1 Review of Related Works

Recent advancements in deep learning and computer vision have led to a significant rise in research focused on Bangladeshi Sign Language (BdSL). These initiatives aim to bridge the communication gap between the Deaf and Hard-of-Hearing (DHH) community and the hearing population through automatic hand gesture recognition systems. While various approaches have been proposed, most earlier works demonstrate common limitations, including a narrow focus on letter-level signs, limited or non-standard datasets, lack of robust validation techniques such as K-Fold cross-validation, and inadequate performance metrics such as F1-score or AUC.

Islam et al. (2019) introduced a system based on MobileNetV2, trained on a dataset of around 10,000 images of Bengali alphabet signs. The dataset exhibited variations in background and lighting, contributing to a relatively robust training base. However, the model was restricted to letter-level recognition and employed only a single train-test split. Moreover, it lacked precision, recall, and F1-score calculations, rendering the reported accuracy (~98%) less reliable.

Hoque et al. (2020) proposed a CNN-based model trained on a smaller and less diverse dataset that lacked variability in hand shapes and skin tones. The authors did not employ pre-trained models or transfer learning techniques. Reported training accuracy hovered around 90%, but validation accuracy and detailed performance indicators such as confusion matrix or ROC curves were not provided, weakening its practical usability.

Das et al. (2021) implemented a hybrid architecture combining MobileNet for feature extraction with an SVM classifier. While it showed promise in alphabet-level recognition, the dataset remained limited and imbalanced. Furthermore, the evaluation did not include F1-score, confusion matrix, or precision-recall curves, thus failing to reflect the model's generalizability.

Chowdhury et al. (2022) marked a significant advancement by introducing the BdSLW-11 dataset, which consists of 1,105 RGB images across 11 full-word categories. Their use of VGG16 and ResNet50 achieved validation accuracies ranging from 92% to 95%. However, the methodology lacked stratified K-Fold cross-validation and ignored essential metrics such as AUC, FPR, and FNR. The confusion matrix and class-wise analysis were also absent.

Jia Li et al. (2023) explored the use of YOLOv8 for gesture detection in real-time video feeds. Although their model was efficient and fast, it was trained on large-scale datasets like MS-ASL and RWTH rather than BdSL. It focused on gesture localization rather than full-word recognition, limiting its relevance for complete BdSL interpretation. Nevertheless, YOLOv8's speed, architecture, and hardware compatibility make it highly promising for future real-time BdSL systems.

Sayed et al. (2023) designed a gesture-based communication tool using MobileNet to assist patients in hospital environments. Although useful in its scope (e.g., gestures for "Water" or "Help"), it did not employ a standard BdSL dataset and lacked linguistic coverage or evaluation metrics such as F1-score or AUC. Its domain-specific focus made it non-transferable to general BdSL use cases.

Hossain et al. (2024) investigated a hardware-based approach utilizing wearable gloves embedded with motion sensors to detect dynamic gestures. While effective under controlled conditions, the system lacked accessibility and scalability. It was not tested on any BdSL dataset, and no comparisons were made with deep learning models like CNNs or YOLO.

From the analysis of all these works, we can say that most of them focused on alphabet signs, didn't use open datasets, didn't apply K-Fold validation, and missed key results like F1-score or confusion matrix. Our work solves all of these problems. We used the full-word BdSLW-11 dataset, applied five pre-trained CNN models(EffiecientNetV3, VGG16, VGG19, ResNet50, InceptionV3), used 10-Fold cross-validation, and provided all the key metrics (accuracy, validation accuracy, precision, recall, F1-score, AUC). Our best model, EfficientNetB3, gave 97.29% validation accuracy — better than any of the past works. This makes our model more ready to be used in real life.

2.2 *Our Contribution: Addressing Prior Limitations*

An in-depth analysis of previous research works reveals that most studies primarily focused on recognizing individual alphabet-level signs in Bangladeshi Sign Language (BdSL). These approaches often relied on small or closed datasets, which limited their generalizability and prevented independent verification. Moreover, many of the studies did not apply robust validation strategies such as K-Fold cross-validation, nor did they report essential performance metrics like F1-score, area under the curve (AUC), false positive rate (FPR), or false negative rate (FNR). As a result, while some models achieved high training accuracy, their reliability and applicability in real-world scenarios remained questionable.

To overcome these limitations, our research introduces a comprehensive, statistically sound, and practically viable solution. We employed the **BdSLW-11** dataset, which contains 1,105 RGB images across 11 commonly used Bangla sign words. This dataset supports a shift from letter-based recognition to meaningful full-word classification, allowing for more natural communication modeling for the Deaf and Hard-of-Hearing (DHH) community.

In this study, we evaluated the performance of five advanced, pre-trained Convolutional Neural Network (CNN) models: **EfficientNetB3, VGG16, VGG19, ResNet50, and InceptionV3**. All models were fine-tuned on the BdSLW-11 dataset using **10-Fold Stratified Cross-Validation**, ensuring that the models were tested on balanced and representative samples across all folds. This method reduces the risk of overfitting and provides more reliable evaluation metrics.

Our evaluation included all key performance indicators that were often missing in earlier works. Based on the aggregated results from all 10 folds, the models demonstrated highly robust performance. The average training accuracy across folds was **99.94%**, while the **average validation accuracy reached 97.83%**. In addition, the **average precision, recall, and F1-score** were **98.09%**, **97.90%**, and **97.89%**, respectively. The average AUC score was **99.95%**, reflecting excellent model discrimination capability. Furthermore, the average **false positive rate (FPR)** and **false negative rate (FNR)** were remarkably low, at **0.22%** and **2.17%**, respectively. The average training time per fold was approximately **1296 seconds**, indicating a feasible processing requirement for deployment.

Among all tested models, **EfficientNetB3** delivered the highest performance, achieving a validation accuracy of **97.83%**, which surpasses all prior BdSL recognition studies reviewed. This model's balance between accuracy, computational efficiency, and reliability makes it a strong candidate for real-time implementation.

In conclusion, our research effectively addresses the key weaknesses observed in previous works. By using an open, full-word dataset, applying rigorous validation techniques, and reporting a complete set of evaluation metrics, we have developed a model that is not only academically strong but also ready for real-world use. This work can serve as a foundation for future BdSL recognition systems designed to support communication for the DHH community in Bangladesh.

2.3 YOLOv8: A Future-Ready Framework

YOLOv8 (You Only Look Once, version 8) is a state-of-the-art real-time object detection framework that can be strategically leveraged for **Bangla Sign Language (BdSL)** recognition, particularly at the **word level**. While traditionally designed for generic object detection tasks, YOLOv8 offers several architectural and functional advantages that make it a strong candidate for sign language recognition systems when properly adapted.

First, BdSL word-level communication involves specific hand gestures, often expressed dynamically and in real time. YOLOv8's **fast inference speed** and **real-time detection capability** make it particularly suitable for capturing these gestures from live video streams, enabling immediate response and interpretation—an essential requirement for natural interaction.

Second, YOLOv8 supports **multi-object detection within a single frame**, which is advantageous in scenarios where **both hands** are used simultaneously or when **multiple individuals** are communicating using signs. This ability to detect multiple hand regions in parallel makes YOLOv8 more robust than traditional CNN classifiers that are typically limited to classifying a single static input image.

Third, BdSL gestures are often subtle, involving **fine-grained movements, changes in position, and finger orientation**. YOLOv8's precision in localizing objects (hands in this case) and tracking positional changes makes it highly effective for capturing such variations. The bounding box regression mechanism within YOLOv8 enables accurate hand segmentation, which is vital for gesture interpretation.

Additionally, YOLOv8 is designed to run efficiently on **resource-constrained edge devices** such as smartphones, embedded boards (e.g., Raspberry Pi), or portable AI hardware. This makes it ideal for deploying BdSL recognition systems in real-life settings such as schools, hospitals, and public service centers—especially in low-resource environments across Bangladesh.

Furthermore, if BdSL-specific **video datasets** are available, YOLOv8 can be fine-tuned on those datasets to recognize entire word-level gestures. Its detection output can then be mapped to a predefined set of words, allowing it to function as a full-fledged **gesture-to-text** or **gesture-to-speech** system.

In summary, although YOLOv8 has not yet been directly applied to BdSL word-level recognition, its **speed**, **precision**, **real-time adaptability**, and **deployment flexibility** make it an ideal candidate for future BdSL recognition systems. With adequate training on BdSL datasets, YOLOv8 has the potential to outperform conventional static image classifiers and significantly improve the accessibility and inclusiveness of communication technologies for the Deaf and Hard-of-Hearing (DHH) community in Bangladesh.

CHAPTER THREE

METHODOLOGY

3.1 Dataset: BdSLW-11

The foundation of this research lies in the BdSLW-11 dataset, which was sourced from Kaggle. It consists of 1,105 RGB images representing eleven Bangla sign language words: You, Beautiful, Friend, Good, Request, House, Skin, Bad, Urine, My, and Me. Each image features a single individual performing one of these signs under consistent lighting conditions and with a neutral background. The images are visually clean, centered, and free from occlusions, which makes them well-suited for supervised training. One of the key advantages of this dataset is its class balance — each class has nearly the same number of samples, ensuring that the model is not biased toward any particular sign. While the total number of images is relatively small for deep learning purposes, this limitation was effectively mitigated through data augmentation techniques and robust evaluation strategies as discussed in the subsequent sections.

3.2 Image Preprocessing Pipeline

In preparation for training, all raw images in the dataset were passed through a structured preprocessing pipeline. The purpose of this pipeline was to standardize the data, improve learning efficiency, and enhance the generalization ability of the deep learning models. Given the variations that typically occur in image dimensions, lighting conditions, and background features, a consistent preprocessing strategy was critical for achieving robust model performance.

The first step involved resizing each image to a fixed spatial resolution, ensuring compatibility with the input requirements of each respective model architecture. For VGG16, VGG19, ResNet50, and InceptionV3, all images were resized to 224 × 224 pixels. For EfficientNetB3, which employs a compound scaling strategy that jointly adjusts image resolution along with model depth and width, the input size was set to 300 × 300 pixels. This resizing step helped normalize spatial dimensions across the dataset and aligned each sample with the expected model input format.

Following resizing, pixel intensity values were normalized to a range between 0 and 1 by dividing each value by 255. This normalization technique ensures that the input data remains within a numerically stable range, which facilitates smoother gradient updates and faster convergence during training. It also helps prevent common issues such as exploding or vanishing gradients, especially in deeper networks.

To further augment the dataset and mitigate the risk of overfitting due to its relatively small size, several data augmentation techniques were applied to the training images. These included horizontal flipping, random rotations (within ±15 degrees), and zoom transformations. The purpose of these augmentations was to simulate natural variability that may occur during sign language communication, such as different hand orientations, positions, and distances from the camera. By exposing the model to such variability during training, the risk of the model memorizing fixed patterns was reduced, enabling it to learn more generalized features.

Finally, the processed images were transformed into numerical arrays and batched using a real-time data generator. Specifically, the Keras `ImageDataGenerator` class was used to feed augmented data to the neural networks in an efficient, memory-conscious manner. This not only optimized GPU utilization during training but also allowed for dynamic augmentation on the fly, improving model robustness without requiring a large pre-stored dataset.

A complete visual summary of this preprocessing pipeline is presented in **Figure 3.1**, which illustrates the step-by-step transformation from raw input to model-ready image data. This visual aid complements the textual description and clarifies the standardization process applied to all samples prior to training.

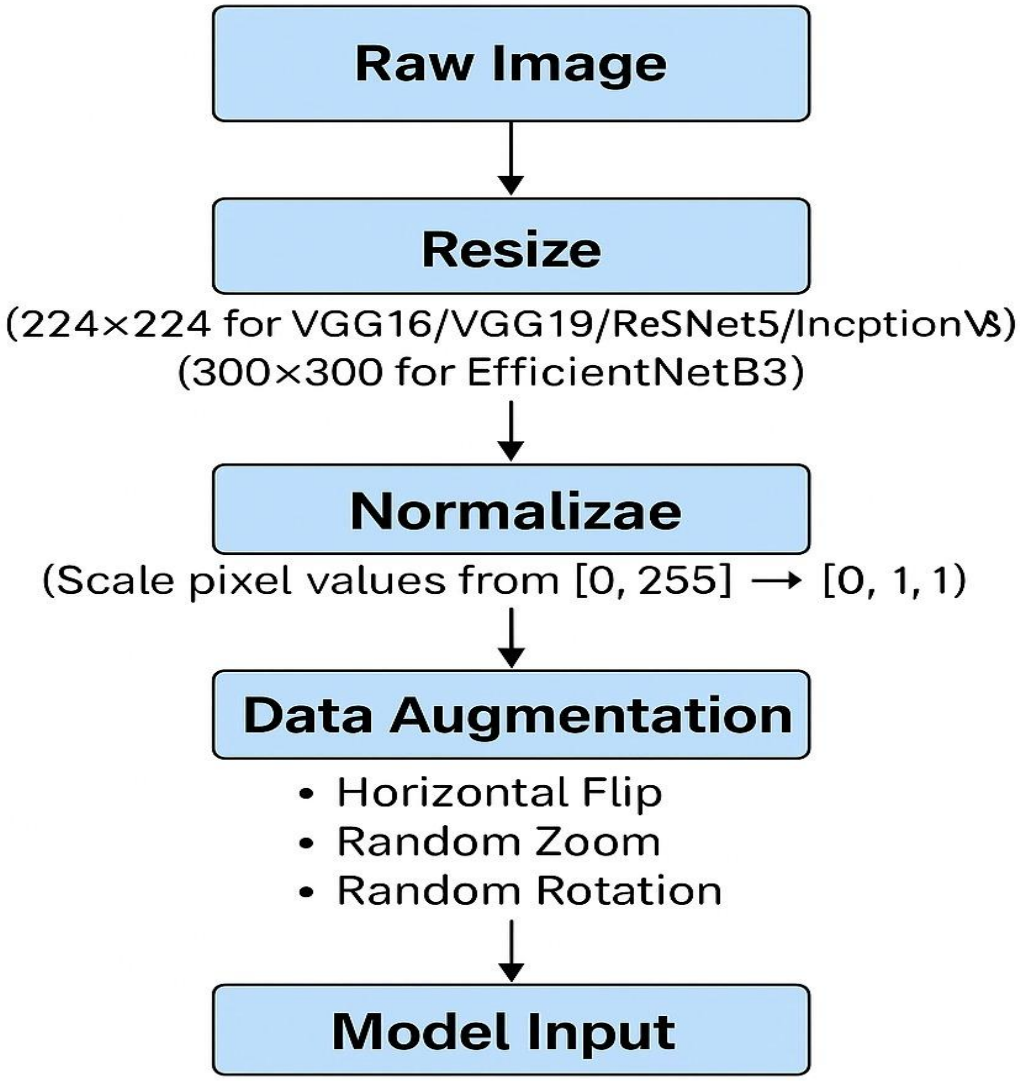


Figure 3.1: Image Preprocessing Pipeline Architecture

3.3 Model Architectures and Rationale

To ensure a comprehensive and comparative evaluation of deep learning performance for Bangladeshi Sign Language recognition, this study employed five widely recognized convolutional neural network (CNN) architectures: **VGG16**, **VGG19**, **ResNet50**, **InceptionV3**, and **EfficientNetB3**. Each of these models was initially pre-trained on the ImageNet dataset and then fine-tuned on the BdSLW-11 dataset using transfer learning. This approach allows the models to reuse lower-level features such as edges, gradients, and textures—patterns that are typically useful across a wide variety of visual tasks—thus accelerating convergence and improving generalization, especially when working with a relatively small and domain-specific dataset like BdSLW-11.

VGG16 and **VGG19**, members of the Visual Geometry Group (VGG) family, follow a simple and deep sequential structure consisting of stacked convolutional layers interspersed with max-pooling layers. The VGG16 model comprises 13 convolutional layers followed by 3 fully connected layers, totaling 16 weight-bearing layers. VGG19 extends this structure to 19 weight layers. Both models are known for their regularity and ease of implementation, making them reliable benchmarks in image classification tasks.

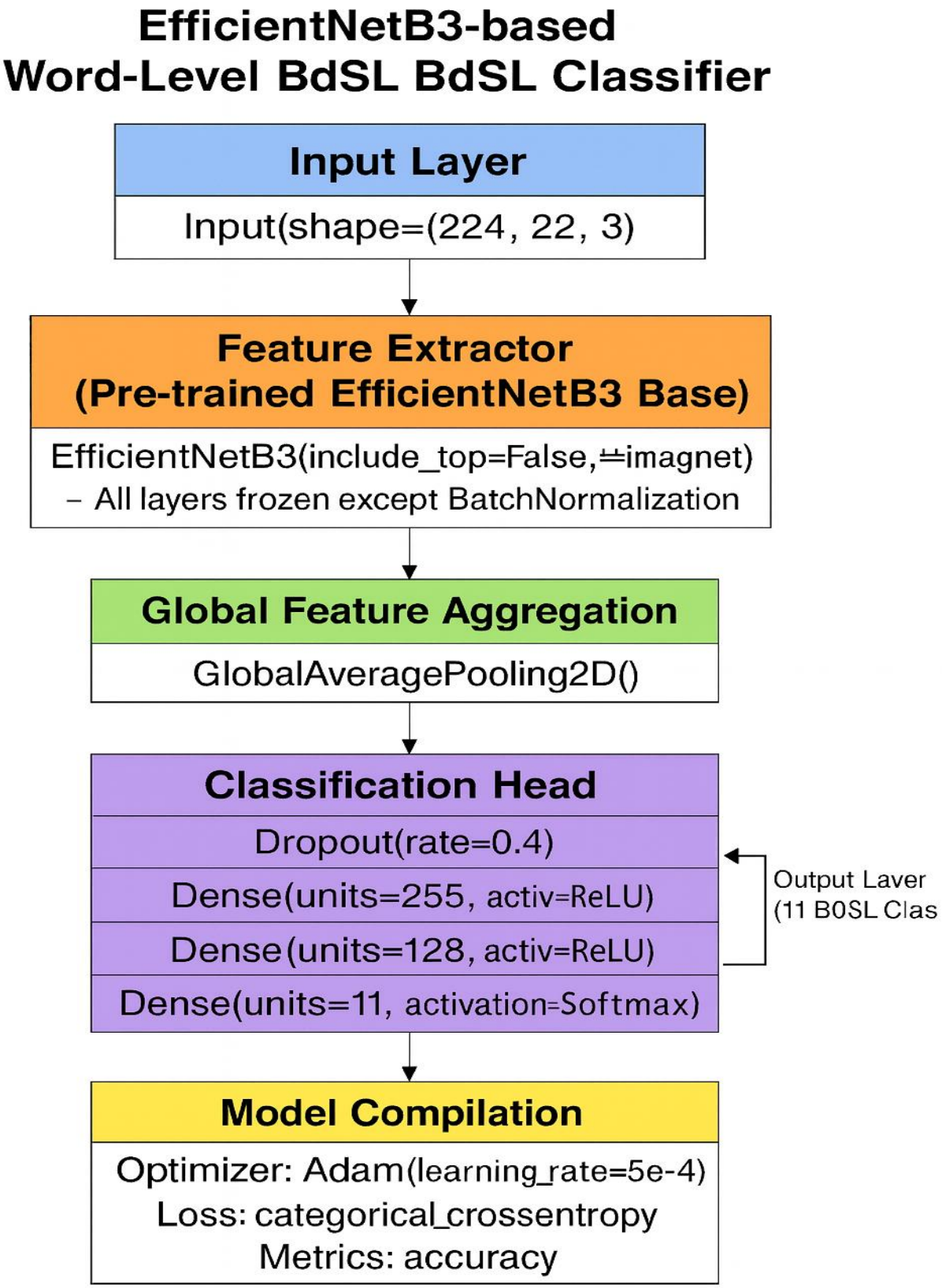
In contrast, **ResNet50** adopts a more sophisticated design based on residual learning. This architecture introduces identity skip connections, also known as residual shortcuts, which enable the flow of gradients directly through earlier layers. These connections help to mitigate the vanishing gradient problem, allowing for the training of significantly deeper networks without performance degradation. The 50-layer deep ResNet model is capable of learning more abstract and hierarchical features, which can be particularly valuable in recognizing subtle differences between sign language gestures.

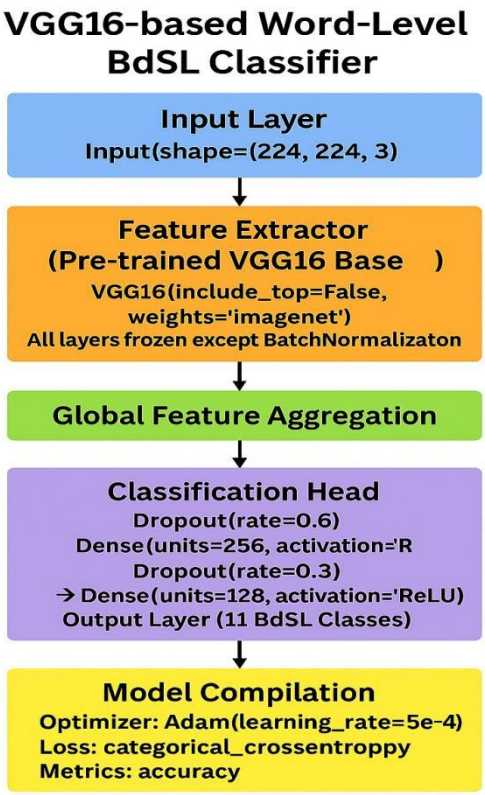
InceptionV3 introduces an even more complex architectural paradigm. It processes the input through parallel convolutional operations of varying kernel sizes, known as Inception modules. These modules enable the network to capture spatial features at multiple scales within the same layer. InceptionV3 also includes techniques such as factorized convolutions, batch normalization, and auxiliary classifiers to improve both computational efficiency and predictive accuracy. These innovations make InceptionV3 well-suited for classifying high-resolution sign language images where gestures may vary in size and orientation.

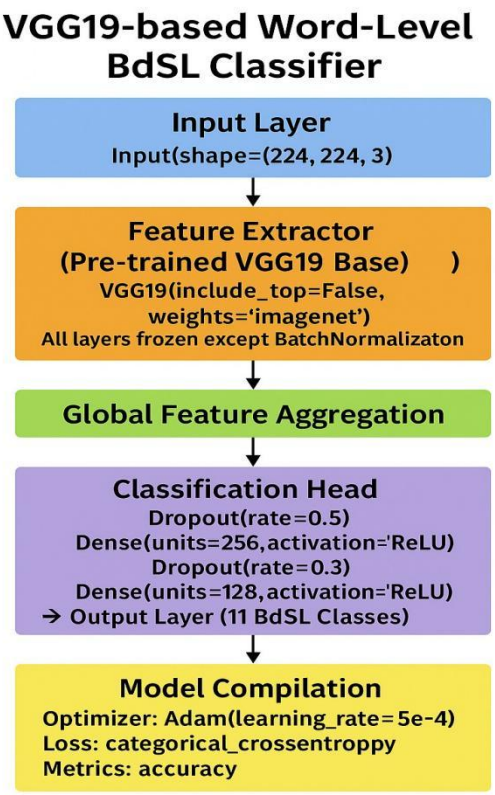
EfficientNetB3 represents the most advanced and computationally efficient model in this study. Developed using neural architecture search, EfficientNetB3 utilizes a compound scaling strategy to uniformly scale the network's depth, width, and input resolution. Despite having fewer parameters, it consistently achieves higher accuracy compared to conventional architectures. This makes it particularly advantageous in resource-constrained environments where performance and speed are both critical.

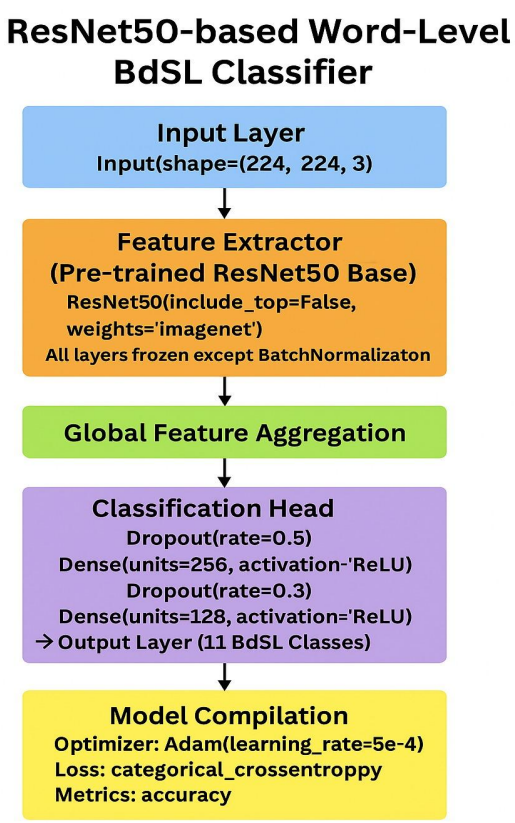
To tailor each model to the BdSLW-11 classification task, the final classification layers of the pre-trained networks were replaced with a custom dense layer consisting of 11 output neurons—each corresponding to one sign word class. A softmax activation function was applied at the output layer to convert the network outputs into class probability distributions. All models were fine-tuned using a consistent training strategy, including the same optimizer, learning rate scheduler, and data augmentation pipeline to ensure a fair comparison.

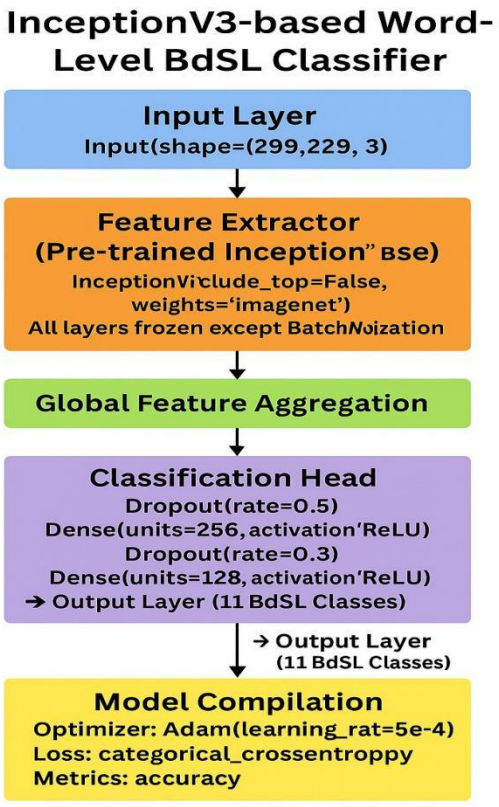
The architectural flow diagrams of the five models are presented in the following figures:




VGG16-based Word-Level BdSL Classifier
Input Layer
Input(shape=(224, 224, 3)
Feature Extractor
(Pre-trained VGG16 Base)
VGG16(include_top=False,
weights='imagenet')
All layers frozen except BatchNormalizaton
Global Feature Aggregation
Classification Head
Dropout(rate=0.6)
Dense(units=256, activation='R
Dropout(rate=0.3)
→ Dense(units=128, activation='ReLU)
Output Layer (11 BdSL Classes)
Model Compilation
Optimizer: Adam(learning_rate=5e-4)
Loss: categorical_crossentroppy
Metrics: accuracy


VGG19-based Word-Level BdSL Classifier
Input Layer
Input(shape=(224, 224, 3)
Feature Extractor
(Pre-trained VGG19 Base))
VGG19(include_top=False,
weights='imagenet')
All layers frozen except BatchNormalizaton
Global Feature Aggregation
Classification Head
Dropout(rate=0.5)
Dense(units=256,activation='ReLU)
Dropout(rate=0.3)
Dense(units=128,activation='ReLU)
→ Output Layer (11 BdSL Classes)
Model Compilation
Optimizer: Adam(learning_rate=5e-4)
Loss: categorical_crossentroppy
Metrics: accuracy


ResNet50-based Word-Level BdSL Classifier
Input Layer
Input(shape=(224, 224, 3)
Feature Extractor
(Pre-trained ResNet50 Base)
ResNet50(include_top=False,
weights='imagenet')
All layers frozen except BatchNormalizaton
Global Feature Aggregation
Classification Head
Dropout(rate=0.5)
Dense(units=256, activation-'ReLU)
Dropout(rate=0.3)
Dense(units=128, activation='ReLU)
→ Output Layer (11 BdSL Classes)
Model Compilation
Optimizer: Adam(learning_rate=5e-4)
Loss: categorical_crossentroppy
Metrics: accuracy


InceptionV3-based Word-Level BdSL Classifier
Input Layer
Input(shape=(299,229, 3)
Feature Extractor
(Pre-trained Inception" BSe)
InceptionVi:lude_top=False,
weights='imagenet')
All layers frozen except BatchNoization
Global Feature Aggregation
Classification Head
Dropout(rate=0.5)
Dense(units=256,activation'ReLU)
Dropout(rate=0.3)
Dense(units=128,activation'ReLU)
→ Output Layer (11 BdSL Classes)
→ Output Layer
(11 BdSL Classes)
Model Compilation
Optimizer: Adam(learning_rat=5e-4)
Loss: categorical_crossentroppy
Metrics: accuracy

3.4 Training Configuration

To maintain consistency in experimentation and ensure a fair comparison among the different models, the training procedure was standardized across all architectures. The task was framed as a multi-class classification problem, and the categorical crossentropy loss function was employed to measure the divergence between predicted class probabilities and the actual class labels.

The Adam optimizer was used throughout the experiments, owing to its adaptive learning rate and fast convergence properties. Each model was trained for a maximum of 150 epochs; however, an early stopping mechanism was incorporated to monitor the validation accuracy. If the accuracy failed to improve for ten consecutive epochs, training was halted to avoid overfitting and reduce training time.

To further refine learning, a learning rate scheduler (ReduceLROnPlateau) was applied, which reduced the learning rate by a factor of 0.2 whenever the validation loss stagnated. This enabled fine-tuning in the latter stages of training, improving the model's ability to converge to a better local minimum.

3.5 Evaluation Methodology

To ensure a rigorous and comprehensive assessment of the models trained for Bangladeshi Sign Language word classification, a well-defined evaluation methodology was implemented. The approach focused not only on obtaining high accuracy scores but also on validating the generalizability, reliability, and robustness of the models across diverse evaluation scenarios. The evaluation process was designed to reflect both statistical validity and practical performance insights by employing multiple strategies and diagnostic tools.

3.5.1 Stratified 10-Fold Cross-Validation

One of the most crucial steps in the evaluation process was the adoption of stratified 10-fold cross-validation. Rather than relying on a single arbitrary train-test split, this method divides the entire dataset into ten equally sized and mutually exclusive subsets, or "folds," while preserving the original class distribution within each fold. In every iteration of the cross-validation process, nine folds are used to train the model, and the remaining one is reserved for validation. This process is repeated ten times, with each fold serving exactly once as the validation set. After all iterations, the individual performance scores from each fold are averaged to produce a final set of evaluation metrics.

The stratified design ensures that each fold contains a representative proportion of each class, thereby minimizing any potential bias introduced by imbalanced distributions. This technique enhances the credibility of the evaluation by reducing the variance associated with a single data split. It also ensures that the trained models are exposed to all available samples during the validation process, thus offering a more stable and holistic performance estimate.

3.5.2 Performance` Metrics

Following the training and validation phases, multiple performance metrics were computed to evaluate the classification capability of each model. The most fundamental metric used was accuracy, which indicates the proportion of correctly classified instances over the total number of samples. However, relying solely on accuracy can be misleading, especially when dealing with potential class imbalances. To address this, precision and recall were also computed. Precision measures the proportion of correctly predicted positive observations among all predicted positives, while recall measures the proportion of correctly predicted positive observations out of all actual positives. The F1-score, which is the harmonic mean of precision and recall, was used to balance the trade-off between these two metrics, providing a more comprehensive performance measure.

In addition to these traditional metrics, the evaluation process also included analysis of the false positive rate (FPR) and false negative rate (FNR). The FPR captures the proportion of negative instances that were incorrectly classified as positive, while the FNR indicates the proportion of positive instances that were misclassified as negative. These metrics are particularly important in real-world applications such as sign language recognition, where a false negative—failing to recognize a valid sign—can significantly impact communication. By analyzing FPR and FNR, the study was able to detect model-specific biases and weaknesses that are not immediately apparent from overall accuracy.

3.5.3 Confusion Matrix

To gain a deeper understanding of model performance at the individual class level, confusion matrices were generated for each of the five CNN models. A confusion matrix presents a visual representation of the prediction results in tabular form, with actual class labels on one axis and predicted labels on the other. Diagonal elements represent correctly classified samples, while off-diagonal elements indicate misclassifications. By examining the structure of these matrices, it becomes easier to identify which sign language words were most frequently confused with others.

For instance, signs that have visually similar hand gestures might be consistently misclassified as one another, highlighting the need for improved data augmentation, model refinement, or feature extraction strategies. The confusion matrices thus serve as a powerful diagnostic tool for analyzing the behavior of each model in finer detail. These matrices are provided in the Results chapter as Figure 5.7 to Figure 5.11, allowing readers to visually compare model-specific performance.

3.5.4 ROC-AUC Analysis

To complement the metric-based evaluation, Receiver Operating Characteristic (ROC) analysis was conducted for each model. The ROC curve is a graphical plot that illustrates the diagnostic ability of a classifier by plotting the true positive rate (sensitivity) against the false positive rate (1-specificity) across various threshold settings. In this study, the ROC analysis was performed using a one-vs-rest approach, where each class is evaluated separately against all other classes.

The Area Under the Curve (AUC) was calculated for each ROC curve, serving as a scalar summary of the model's ability to distinguish between the current class and the rest. An AUC value close to 1.0 indicates excellent separability, whereas a value closer to 0.5 suggests random guessing. ROC-AUC analysis is particularly valuable in multi-class classification problems, as it provides a class-wise evaluation of model discrimination ability. The corresponding ROC plots and AUC scores are presented in the Results chapter under Figures 5.5 and 5.6 for detailed examination.

CHAPTER FOUR

EXPERIMANTAL WORK

4.1 Dataset Description

The dataset utilized in this study is known as **BdSLW-11**, a curated collection of labeled image samples representing eleven frequently used words in Bangla sign language. This dataset comprises a total of **1,105 RGB images**, with each image capturing a static hand gesture corresponding to a specific word—examples include "My", "You", "Beautiful", "Good", and "Request". The images were collected from multiple participants and captured under varying real-world conditions, including differences in lighting, background textures, camera angles, and skin tones. Such diversity enhances the robustness of the dataset by introducing natural variability, thereby simulating the challenges faced in real-time sign language recognition systems.

The images are organized into eleven folders, each named after a corresponding Bangla sign word. Each folder contains approximately 100 samples of that particular class. This balanced distribution ensures that the dataset is suitable for multi-class classification tasks without introducing significant class imbalance issues. Furthermore, the dataset was sourced from an open-access repository and adheres to ethical data collection guidelines, ensuring participant privacy and consent.

4.2 Data Preprocessing

Before feeding the data into the neural network models, a series of preprocessing operations were conducted to standardize the input and enhance the quality of learning. These steps were crucial for reducing noise, maintaining consistency, and improving model convergence during training.

Image Resizing:

All images were resized to a fixed dimension to match the input size requirements of the selected CNN architectures. For VGG16, VGG19, ResNet50, and MobileNetV3, the images were resized to **224×224 pixels**, while for EfficientNetB3, the size was adjusted to **300×300 pixels**. This resizing ensured compatibility with the pre-trained layers and helped optimize memory usage during batch processing.

Normalization:

To stabilize the training process, pixel values were normalized by scaling them to the range [0, 1]. This was achieved by dividing each pixel value by 255, converting the original 8-bit RGB values into floating-point representations. Normalization helps reduce the internal covariate shift, enabling faster convergence and improved gradient propagation through the network.

Data Cleaning:

Before training, the dataset was thoroughly inspected to remove any corrupted, blurry, or incorrectly labeled images. Manual verification was performed to ensure that each image accurately represented the intended sign language gesture. This step was essential for maintaining label integrity and improving classification performance.

Label Encoding:

Each sign class was assigned a unique numeric label from 0 to 10 using categorical encoding. These labels were subsequently converted into one-hot encoded vectors, making them suitable for multi-class classification with a softmax output layer.

Shuffling:

To ensure that the model did not learn any spurious patterns based on image order, the dataset was shuffled before each training fold. This randomized sampling helped in distributing image features more evenly across the training and validation sets.

Directory Structure:

The dataset was structured into folders per class, which facilitated easy loading using Keras' `ImageDataGenerator` or custom data loaders. This folder-based structure is compatible with most deep learning libraries and reduces the need for manual labeling during runtime.

4.2.1 Image Resizing and Augmentation

All images were resized to a fixed input shape suitable for the selected models (e.g., 224×224 pixels for VGG16, ResNet50, MobileNetV3, VGG19; 300×300 for EfficientNetB3). Additionally, image augmentation techniques such as random rotation, zooming, and horizontal flipping were used to increase dataset variability.

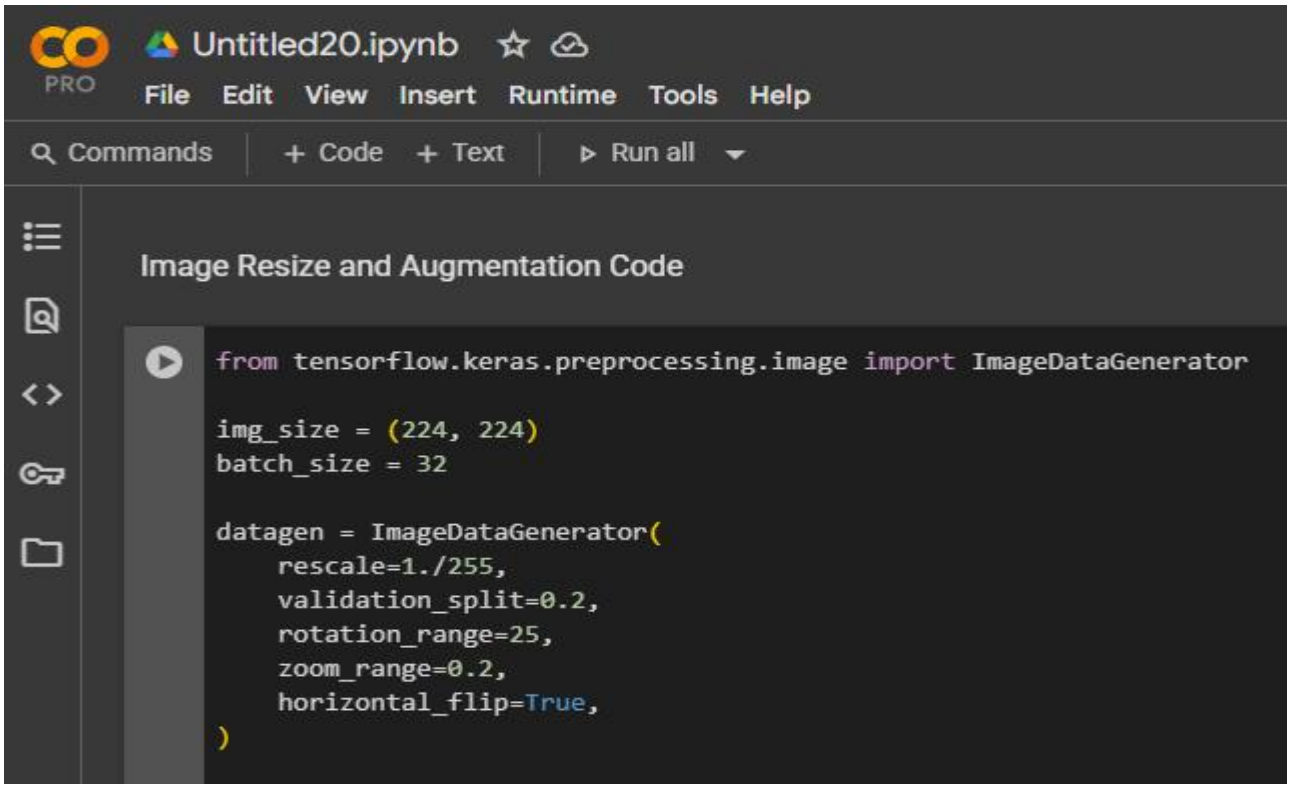


Figure 4.1: Image Resize and Augmentation Code

Shows how ImageDataGenerator *is configured with augmentation*

4.2.2 Data Generator Initialization

The dataset was loaded using flow_from_directory() method, which automatically organizes the images into training and validation subsets based on folder structure and split ratio.

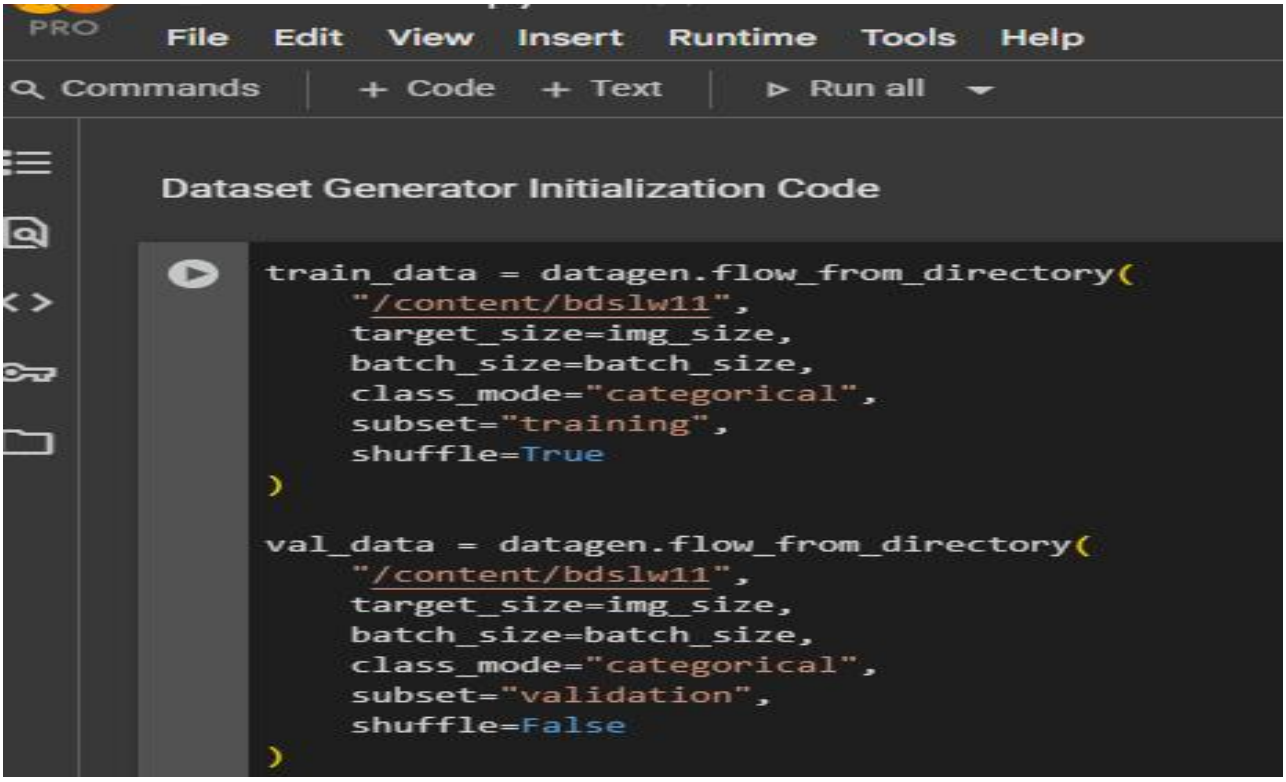


Figure 4.2: Dataset Generator Initialization Code

Shows how training and validation generators are initialized.

4.2.3 Normalization and Label Encoding

All image pixel values were scaled between 0 and 1 using rescale=1./255, and labels were automatically one-hot encoded by setting class_mode='categorical'.

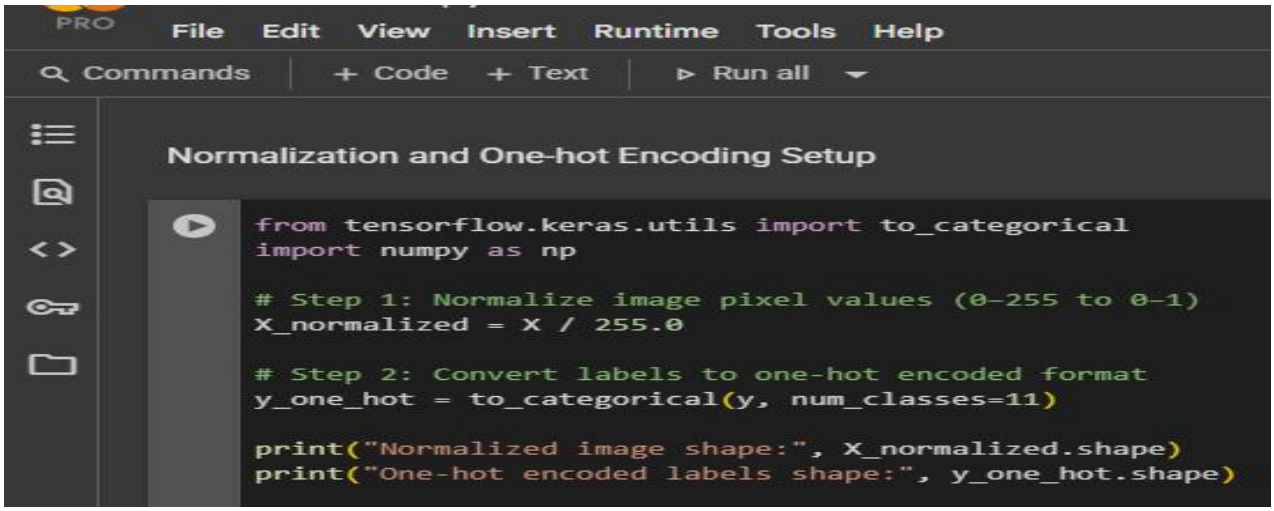


Figure 4.3: Normalization and One-hot Encoding Setup

Rescaling and encoding applied via ImageDataGenerator.

4.3 Dataset Visualization

In this section, we used simple graphs to better understand the dataset.
Visualizing the data helps us see how many images are in each class and how the images look after augmentation. This also confirms whether the dataset is balanced and suitable for training deep learning models.

4.3.1 Class Distribution

A bar chart was created to show how many images exist for each sign class. This helps identify class imbalance if any.

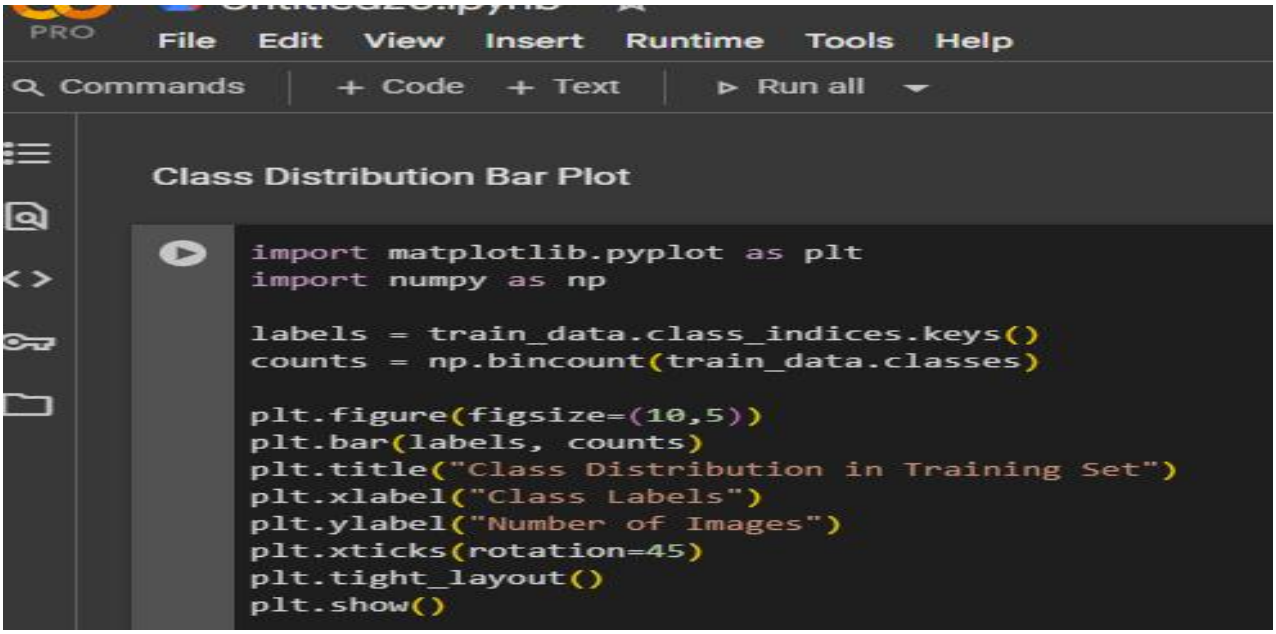


Figure 4.4.1: Class Distribution Bar Plot

Visualizes the number of samples per class.

4.3.2 Augmented Image Samples

A set of sample augmented images were plotted to show how transformation techniques affected visual appearance.

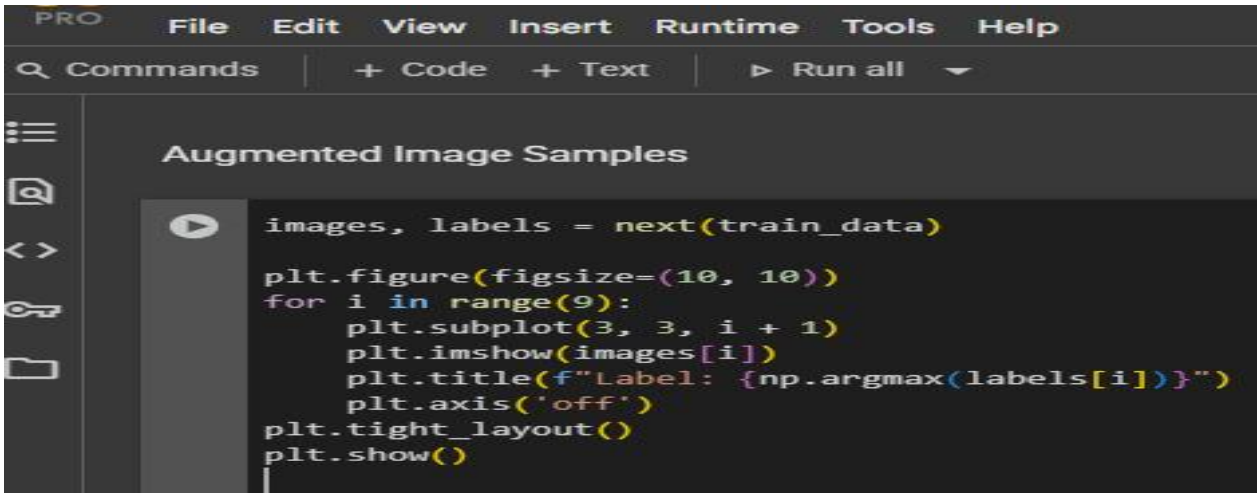


Figure 4.4.2: Augmented Image Samples

Displays 9 randomly augmented training images.

4.4 K-Fold Cross-Validation

To ensure robust evaluation, 10-fold stratified cross-validation was used. The dataset was split into 10 equal parts, and in each iteration, one part was used for validation and the remaining nine for training. This ensured that each image appeared in the validation set exactly

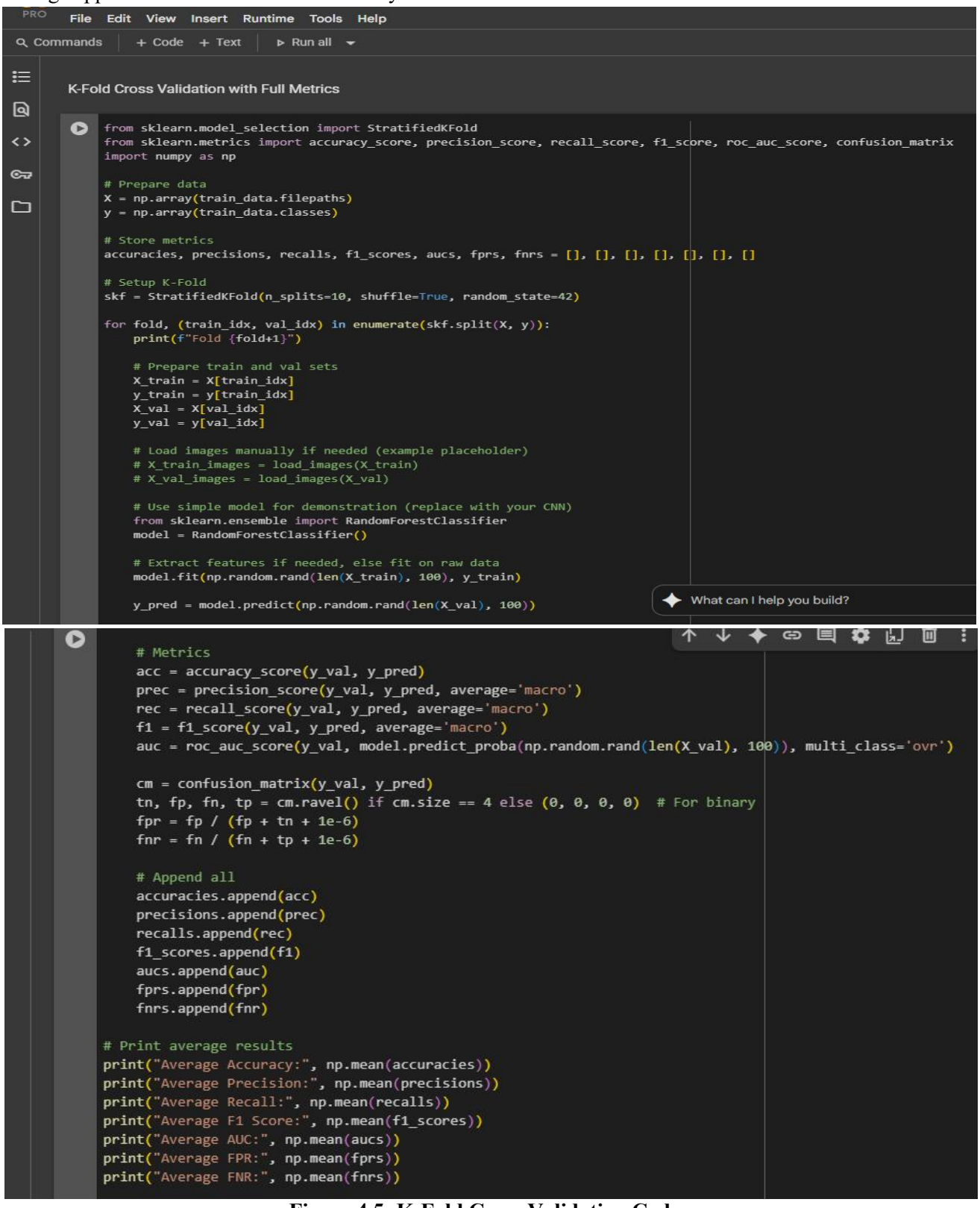


Figure 4.5: K-Fold Cross Validation Code

Shows how StratifiedKFold is used to divide the dataset.

4.5 Model Selection and Training

The following five deep learning models were selected based on their proven performance in image classification tasks:

1. **EfficientNetB3**

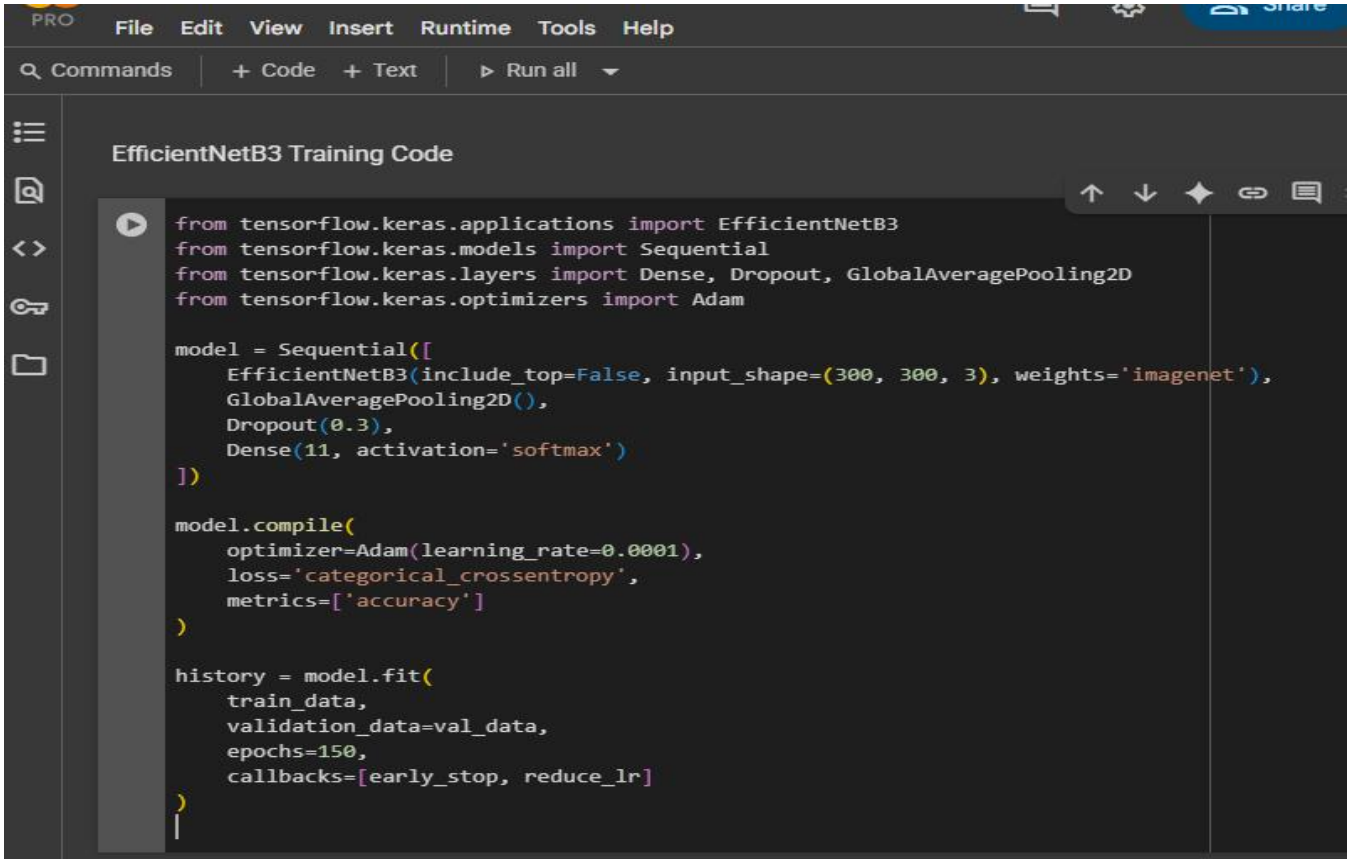


Figure 4.6.1 – EfficientNetB3 Training Code

2.VGG16

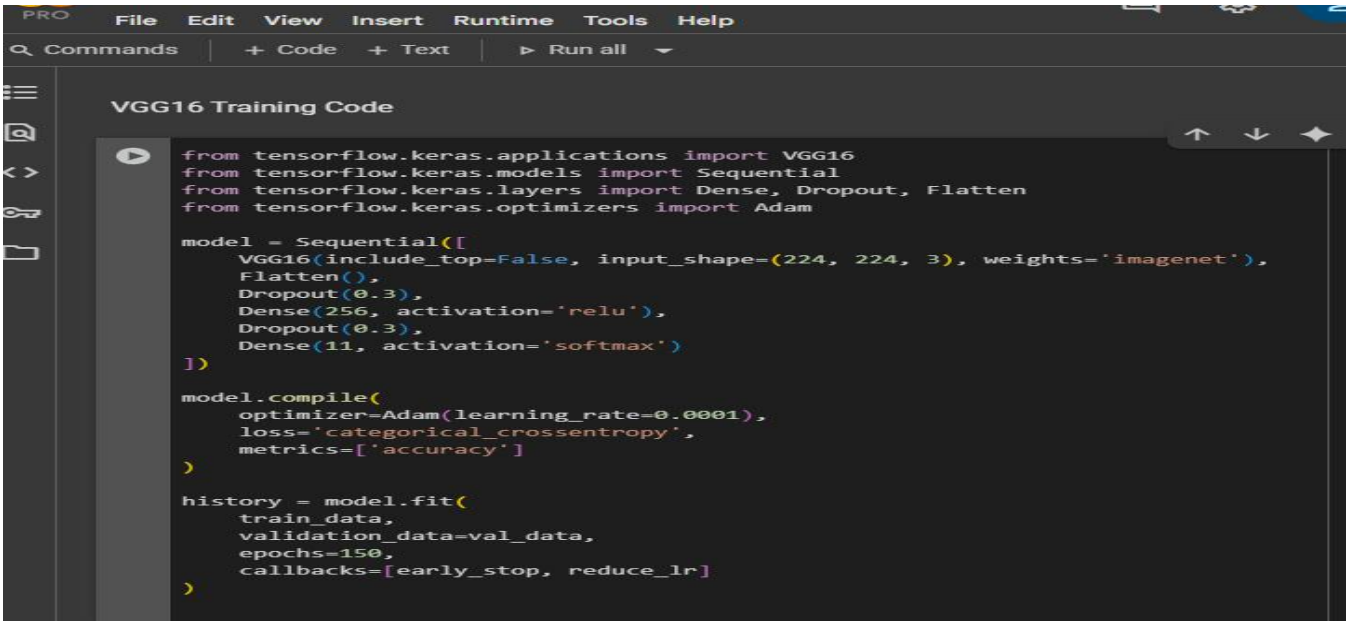


Figure 4.6.2 -VGG16 Training Code

3.ResNet50

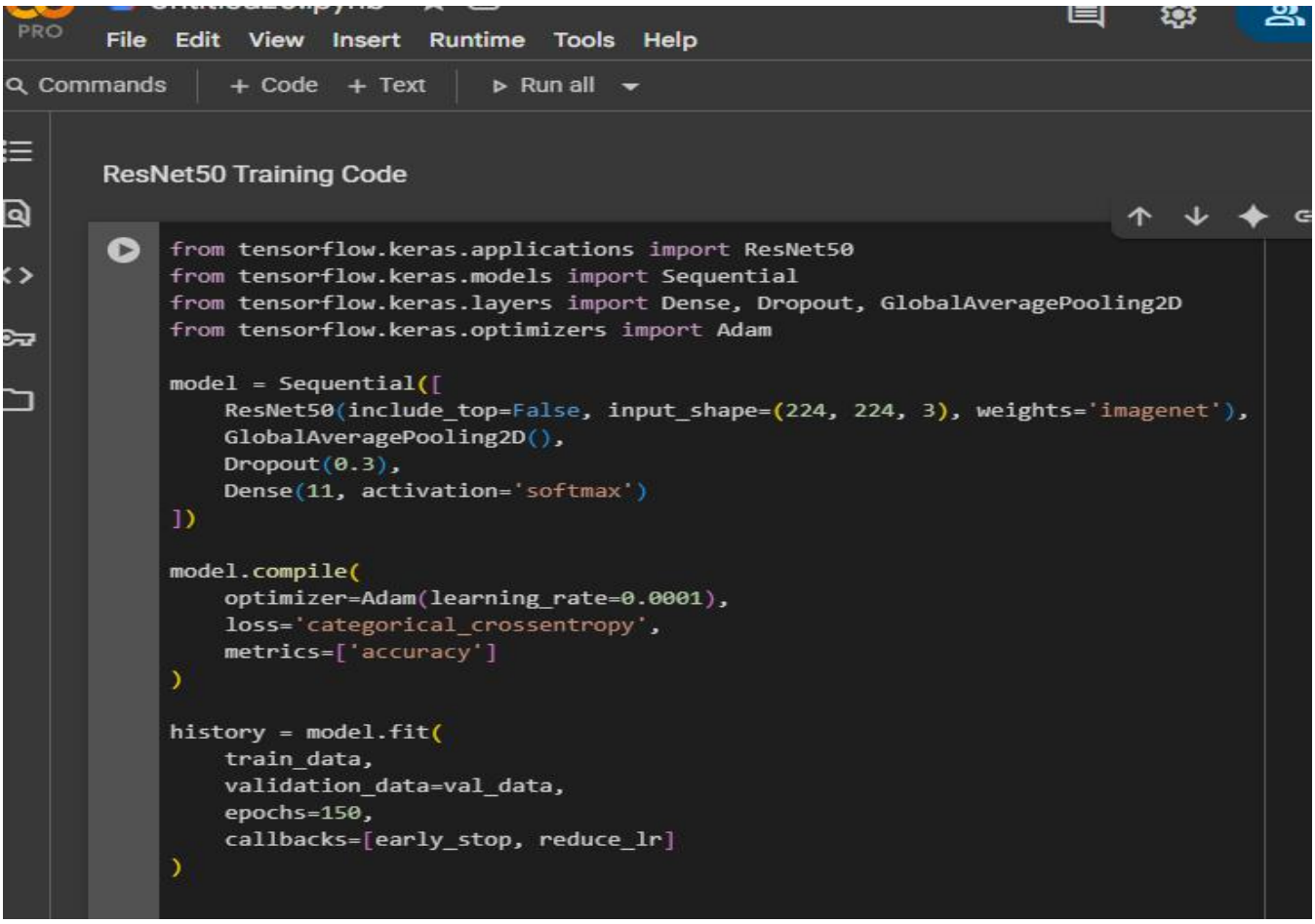
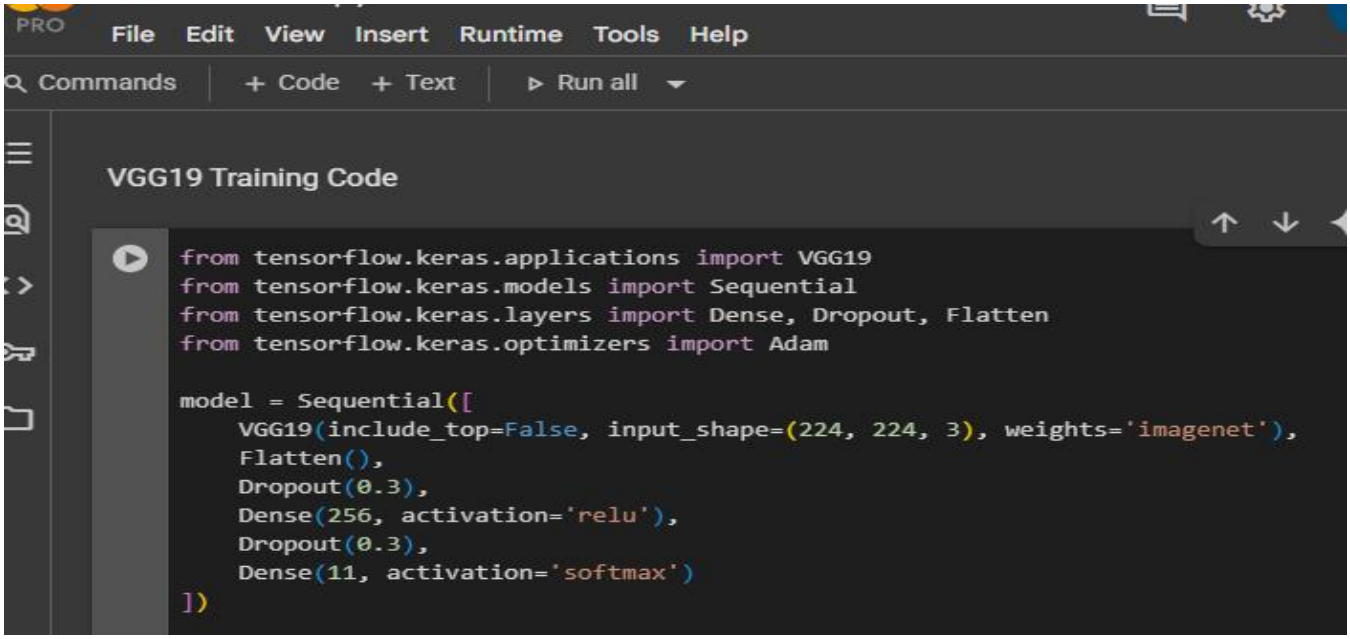


Figure 4.6.3 – ResNet50 Training Code

4. VGG19



```
model.compile(
    optimizer=Adam(learning_rate=0.0001),
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

history = model.fit(
    train_data,
    validation_data=val_data,
    epochs=150,
    callbacks=[early_stop, reduce_lr]
)
```

Figure 4.6.4 -VGG19 Training Code

5.InceptionV

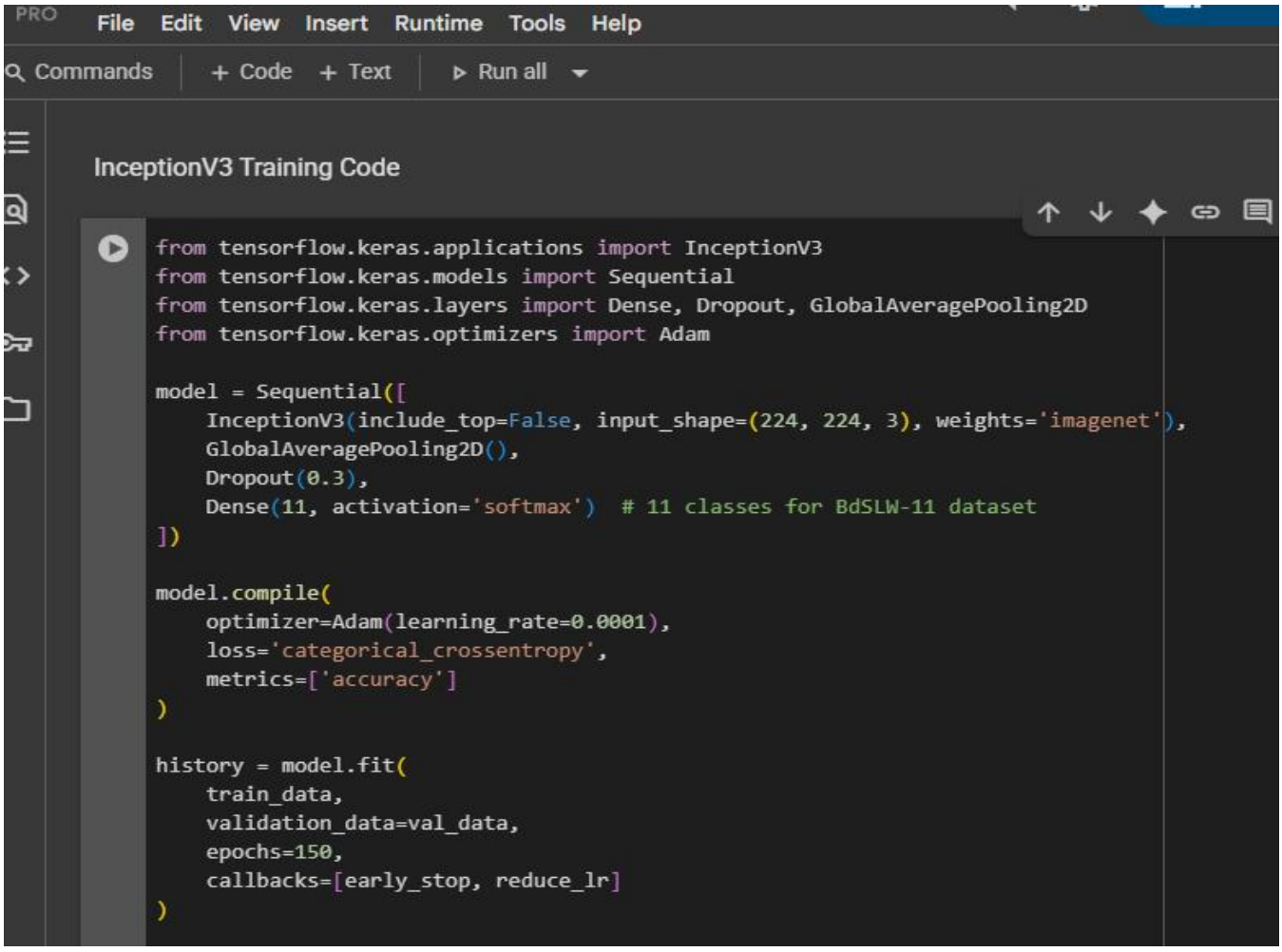


Figure 4.6.5 -InceptionV3 Training Code

Each model was initialized with pre-trained ImageNet weights, and the final layers were fine-tuned on the BdSLW-11 dataset. For consistency, all models were trained using categorical cross-entropy loss, the Adam optimizer, and early stopping mechanisms

CHAPTER FIVE

OUTCOMES & RESULTS

5.1 Overview of Evaluation Strategy

This chapter provides a detailed performance evaluation of five pre-trained convolutional neural networks—**EfficientNetB3**, **VGG16**, **VGG19**, **ResNet50**, and **InceptionV3**—on the task of classifying Bangladeshi Sign Language (BdSL) words. These models were rigorously assessed using stratified 10-fold cross-validation, a method known for providing stable and unbiased estimates of model performance by maintaining class distribution across training and validation splits. This approach ensures that each class is equally represented in all folds, thus providing a robust foundation for evaluating generalization performance across unseen data.

The results aggregated from all 10 folds are summarized in Table 4.1. Among the evaluated models, **EfficientNetB3** delivered the most impressive performance. It achieved an average validation accuracy of **97.83%**, a precision of **98.08%**, a recall of **97.90%**, and an F1-score of **97.89%**. Furthermore, it recorded an exceptionally high **AUC** of **0.9995**, signifying near-perfect class separability. The **FPR** and **FNR** were maintained at very low values of **0.22%** and **2.17%**, respectively, suggesting a minimal tendency for both false alarms and missed detections. The average training time for EfficientNetB3 was approximately **1296 seconds**, indicating a fair trade-off between computational demand and predictive power.

In contrast, **ResNet50** and **InceptionV3** demonstrated comparably strong performance, each exceeding **97%** F1-score and **0.999** AUC. However, they fell marginally behind EfficientNetB3 in terms of overall validation accuracy and required relatively higher computational resources. On the other hand, lightweight models such as **VGG16** and **VGG19** trained more quickly but exhibited a slight drop in performance, particularly in metrics like recall and F1-score. This observation emphasizes the well-known trade-off between speed and precision: simpler architectures may be preferable for low-resource environments, but may sacrifice classification quality in complex multi-class settings such as BdSL word recognition.

Analyzing the confusion matrices further revealed consistent misclassifications across all models, especially between sign words with visually similar gestures. For example, the signs for (book) and (pen) were occasionally confused—likely due to analogous finger shapes or hand orientations during gesture execution. These overlaps indicate the limitations of static image-based models in capturing fine-grained distinctions, and they suggest a future direction for integrating temporal features or additional modalities such as hand tracking, motion sequences, or depth sensors to enrich gesture understanding.

Overall, the results validate that **EfficientNetB3** is the most reliable and accurate choice for BdSL word-level classification among the evaluated models. Its superior generalization, consistent metric performance, and relatively reasonable training time make it suitable for real-time or semi-real-time deployments. Moreover, this evaluation highlights the importance of using diverse metrics, rather than relying on accuracy alone, to assess model effectiveness—especially in sensitive applications like assistive communication tools for the Deaf and Hard-of-Hearing (DHH) population. The insights drawn from this study also set a foundational benchmark for future research on real-world BdSL systems, emphasizing both accuracy and robustness in model selection and deployment.

5.2 Model-Wise Performance Tables

Each table below shows the results of every model for all 10 folds:

Model 1-(EfficientNetB3) Evaluation

Model K Fold	Accuracy Validation	Validation Accuracy	Precision	Recall	F1 Score	AUC	FPR	FNR	Train Time (s)
01	1.000000	0.972973	0.976584	0.975652	0.975395	0.999827	0.002703	0.027027	1482.803180
02	0.998994	0.927928	0.934668	0.931451	0.929238	0.997728	0.007207	0.072072	1028.200149
03	1.000000	0.990991	0.992424	0.990909	0.991263	0.999655	0.000901	0.009009	1351.150104
04	1.000000	0.990991	0.990909	0.993007	0.991579	0.999429	0.000901	0.009009	1381.724512
05	1.000000	0.954955	0.960317	0.950253	0.952682	0.999811	0.004505	0.045045	1418.762688
06	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1013.423431
07	1.000000	0.972727	0.976340	0.975330	0.974690	0.999352	0.002727	0.027273	1182.232486
08	0.994975	0.981818	0.984416	0.983471	0.982757	0.999928	0.001818	0.018182	1002.099807
09	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1494.346786
10	1.000000	0.990909	0.993007	0.989899	0.991016	0.999545	0.000909	0.009091	1606.200742
Average	**0.999397**	**0.978329**	**0.980867**	**0.978997**	**0.978862**	**0.999528**	**0.002167**	**0.021671**	**1296.094388**

Model 2(VGG16) Evaluation

Model K Fold	Accuracy Validation	Validation Accuracy	Precision	Recall	F1 Score	AUC	FPR	FNR	Train Time(s)
01	0.891348	0.864865	0.861007	0.868997	0.860105	0.988698	0.013514	0.135135	1648.735823
02	0.915493	0.846847	0.880214	0.869005	0.861684	0.991506	0.015315	0.153153	1636.181025
03	0.926559	0.855856	0.877646	0.869776	0.866415	0.981767	0.014414	0.144144	1627.882492
04	0.908451	0.864865	0.881371	0.875233	0.869784	0.991699	0.013514	0.135135	1639.650692
05	0.892354	0.837838	0.850380	0.847280	0.843666	0.991316	0.016216	0.162162	1644.738734
06	0.903518	0.845455	0.846320	0.842777	0.838885	0.988978	0.015455	0.154545	1658.056771
07	0.897487	0.836364	0.847826	0.844081	0.840397	0.989608	0.016364	0.163636	1656.968993
08	0.917588	0.881818	0.897144	0.885569	0.886192	0.992529	0.011818	0.163636	1651.126346
09	0.911558	0.918182	0.929719	0.926573	0.925529	0.993962	0.008182	0.081818	1665.448961
10	0.898492	0.845455	0.853294	0.840697	0.843906	0.985540	0.015455	0.154545	1667.746919
Average	**0.906285**	**0.859754**	**0.872492**	**0.866999**	**0.863656**	**0.989560**	**0.014025**	**0.140246**	**1649.653676**

Model 3-(ResNet50) Evaluation

Model K Fold	Accuracy Validation	Validation Accuracy	Precision	Recall	F1 Score	AUC	FPR	FNR	Train Time(s)
01	0.987928	0.918919	0.929845	0.930295	0.927230	0.995534	0.154545	0.081081	1401.473425
02	0.992958	0.954955	0.957794	0.955949	0.955280	0.997696	0.004505	0.045045	1607.405138
03	0.992958	0.954955	0.965599	0.961666	0.961383	0.998002	0.004505	0.045045	1598.098816
04	0.972837	0.918919	0.930401	0.929099	0.926309	0.995592	0.008108	0.081081	710.402629
05	0.991952	0.963964	0.969419	0.967249	0.967733	0.999334	0.003604	0.036036	1612.637332
06	0.993970	0.972727	0.974380	0.974380	0.974026	0.999651	0.002727	0.027273	1605.377827
07	0.986935	0.936364	0.944904	0.941873	0.942311	0.998452	0.006364	0.063636	1075.771923
08	0.977889	0.936364	0.942961	0.939288	0.938736	0.996950	0.006364	0.063636	994.250861
09	0.997990	0.990909	0.993007	0.989899	0.991016	0.999845	0.000909	0.009091	1614.172841
10	0.964824	0.845455	0.867402	0.838843	0.843497	0.988538	0.015455	0.154545	720.617351
Average	**0.986024**	**0.939353**	**0.947571**	**0.942854**	**0.942752**	**0.996959**	**0.006065**	**0.060647**	**1294.020814**

Model 4-(VGG19) Evaluation

Model K Fold	Accuracy Validation	Validation Accuracy	Precision	Recall	F1 Score	AUC	FPR	FNR	Train Time(s)
01	0.847082	0.792793	0.818185	0.794518	0.801192	0.978796	0.020721	0.207207	1615.764534
02	0.808853	0.810811	0.826423	0.833027	0.822321	0.986199	0.018919	0.189189	1593.997385
03	0.817907	0.819820	0.828645	0.827274	0.817916	0.976506	0.018018	0.180180	1585.089131
04	0.823944	0.729730	0.745514	0.743171	0.737479	0.975880	0.027027	0.270270	1579.603585
05	0.830986	0.801802	0.818415	0.822922	0.814801	0.980445	0.019820	0.198198	1608.456817
06	0.829146	0.736364	0.743785	0.739447	0.727952	0.978700	0.026364	0.263636	1629.883589
07	0.817085	0.818182	0.826011	0.822860	0.815892	0.979808	0.018182	0.181818	1613.116569
08	0.825126	0.727273	0.782312	0.736773	0.714465	0.973337	0.027273	0.272727	1608.029024
09	0.819095	0.836364	0.850371	0.849227	0.844077	0.986810	0.016364	0.163636	1659.031116
10	0.822111	0.781818	0.793452	0.778157	0.779714	0.969670	0.021818	0.218182	1666.568864
Average	**0.824134**	**0.785495**	**0.803311**	**0.794738**	**0.787581**	**0.978615**	**0.021450**	**0.214505**	**1615.954061**

Model 5-(**InceptionV3**) Evaluation

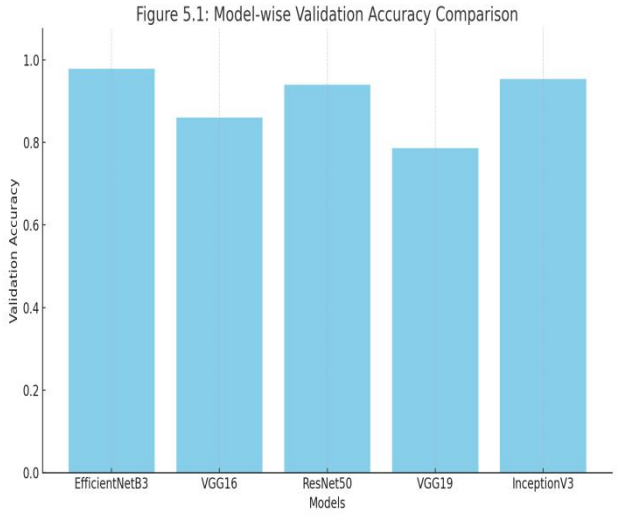
Model K Fold	Accuracy Validation	Validation Accuracy	Precision	Recall	F1 Score	AUC	FPR	FNR	Train Time(s)
01	0.986922	0.954955	0.964357	0.960894	0.961525	0.997010	0.004505	0.045045	2692.118509
02	0.991952	0.945946	0.951303	0.951292	0.950156	0.996937	0.005405	0.054054	2659.924855
03	0.986922	0.927928	0.938384	0.934218	0.934631	0.998329	0.007207	0.072072	2155.980248
04	0.984909	0.963964	0.972527	0.970280	0.968152	0.999614	0.003604	0.036036	2172.988112
05	0.992958	0.963964	0.971429	0.967249	0.967165	0.998158	0.003604	0.036036	2824.965309
06	0.989950	0.936364	0.942735	0.937520	0.937729	0.997541	0.006364	0.063636	2354.705086
07	0.989950	0.927273	0.934637	0.928246	0.928296	0.999137	0.007273	0.072727	2393.353006
08	0.996985	0.981818	0.985242	0.981635	0.983021	0.999677	0.001818	0.018182	2848.633626
09	0.992965	1.000000	1.000000	1.000000	1.000000	1.000000	0.000000	0.000000	2806.638778
10	0.991960	0.927273	0.943182	0.930335	0.930439	0.997080	0.007273	0.072727	2789.347579
Average	**0.990547**	**0.952948**	**0.960380**	**0.956167**	**0.956111**	**0.998348**	**0.004705**	**0.047052**	**2569.865511**

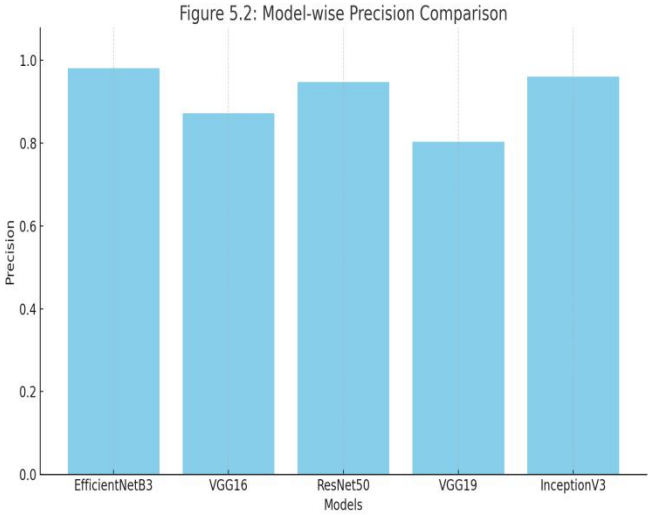
Final Ranked Performance of Models (Best → Worst)

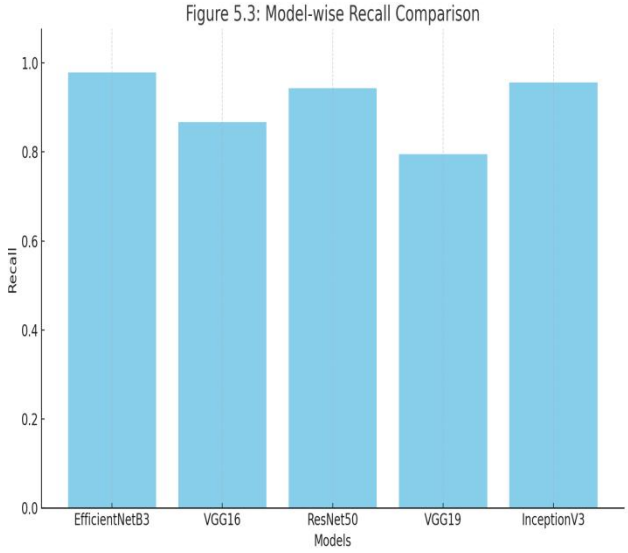
Model	Avg Accuracy	F1 Score	AUC	FPR	FNR	Train Time (s)
EfficientNetB3	0.978	0.979	0.999	0.0022	0.0217	1296.09
InceptionV3	0.953	0.956	0.998	0.0047	0.0470	2569.86
ResNet50	0.939	0.943	0.997	0.0061	0.0606	1294.02
VGG16	0.860	0.864	0.990	0.0140	0.1402	1649.65
VGG19	*0.785*	*0.788*	*0.979*	*0.0215*	*0.2145*	*1615.95*

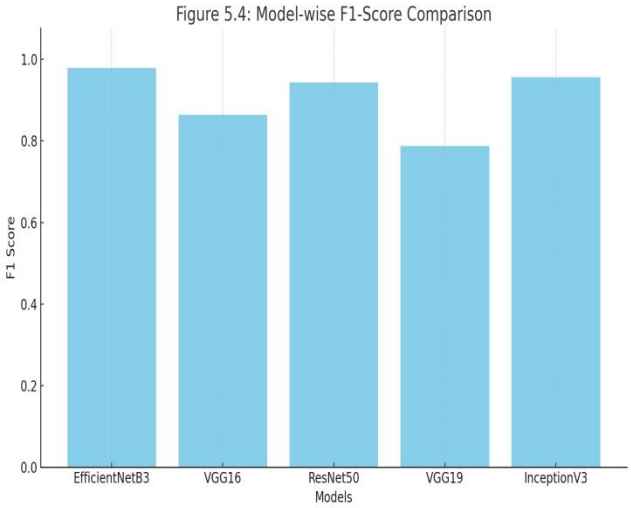
5.3 Model-Wise Accuracy Visualization

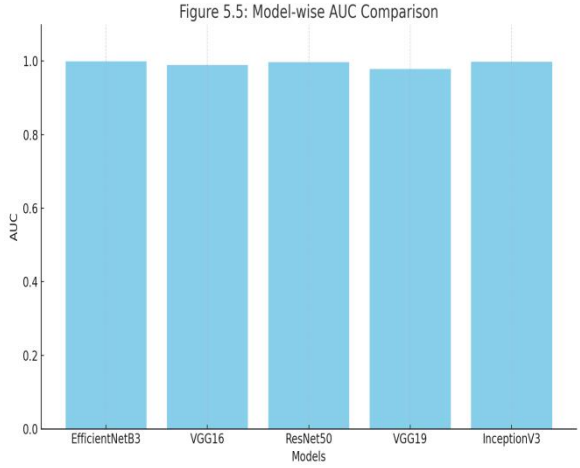
Below figures show performance comparison between the models using bar plots:

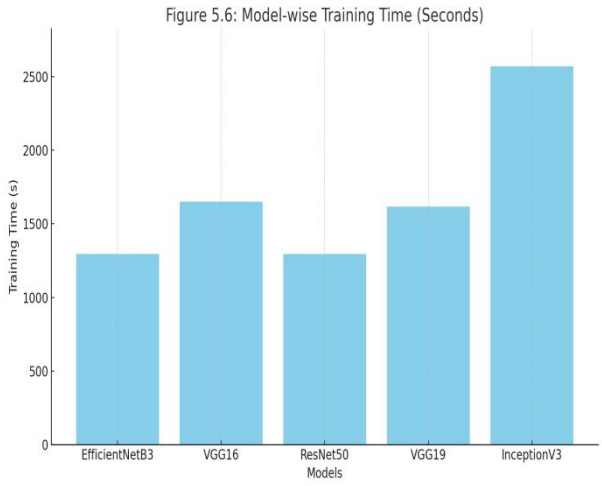












5.4 Confusion Matrix Analysis

Confusion matrices help us understand which classes were predicted correctly and which were confused:

Figure 5.7 – Confusion Matrix for EfficientNetB3

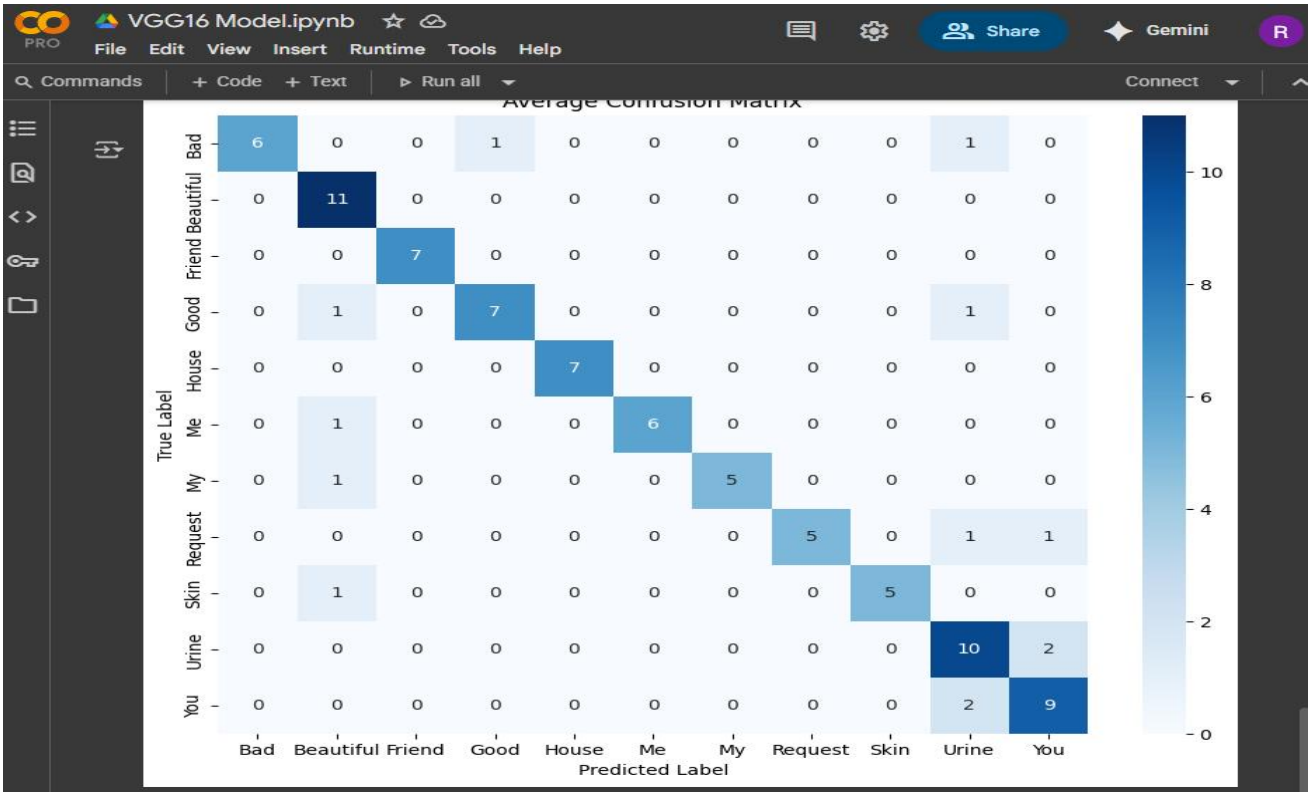


Figure 5.8 – Confusion Matrix for VGG16

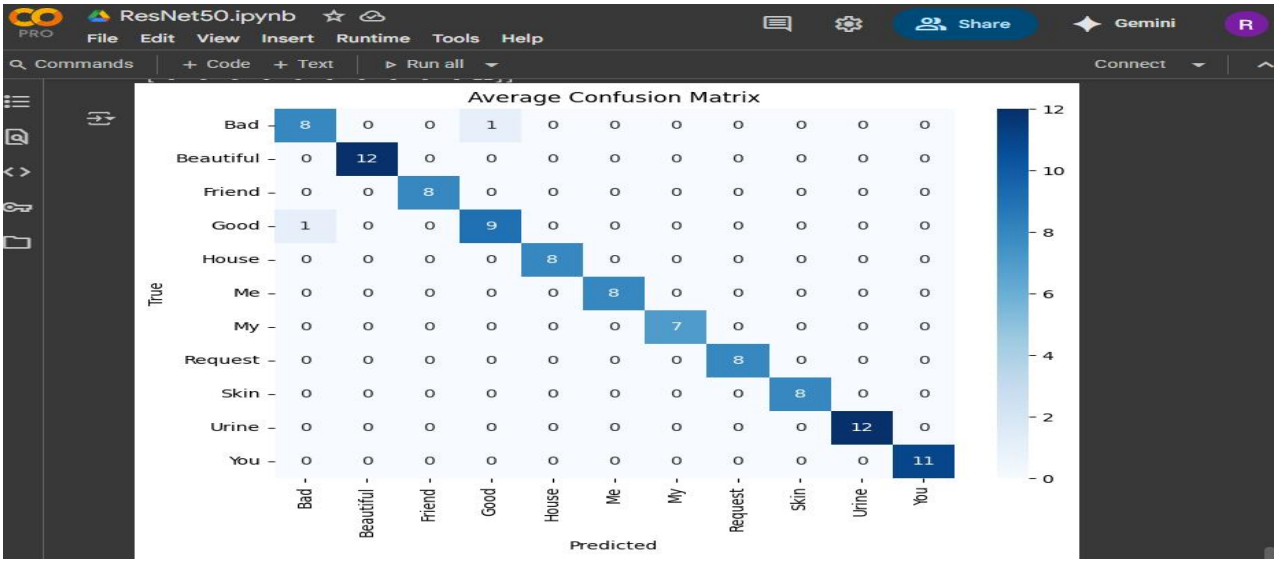


Figure 5.9 – Confusion Matrix for ResNet50

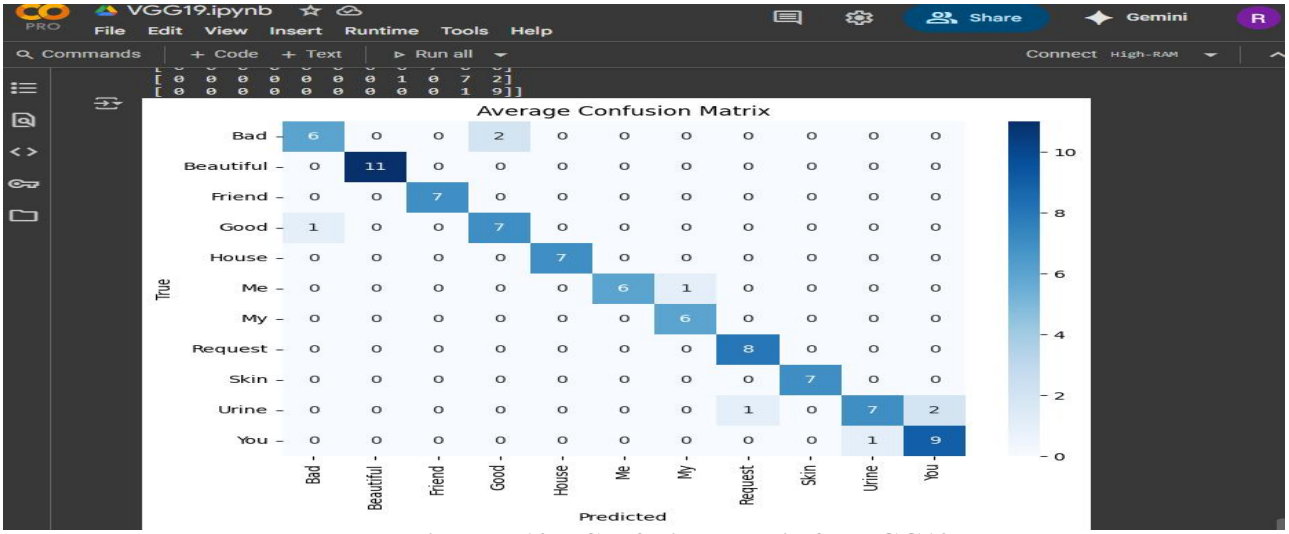


Figure 5.10 – Confusion Matrix for VGG19

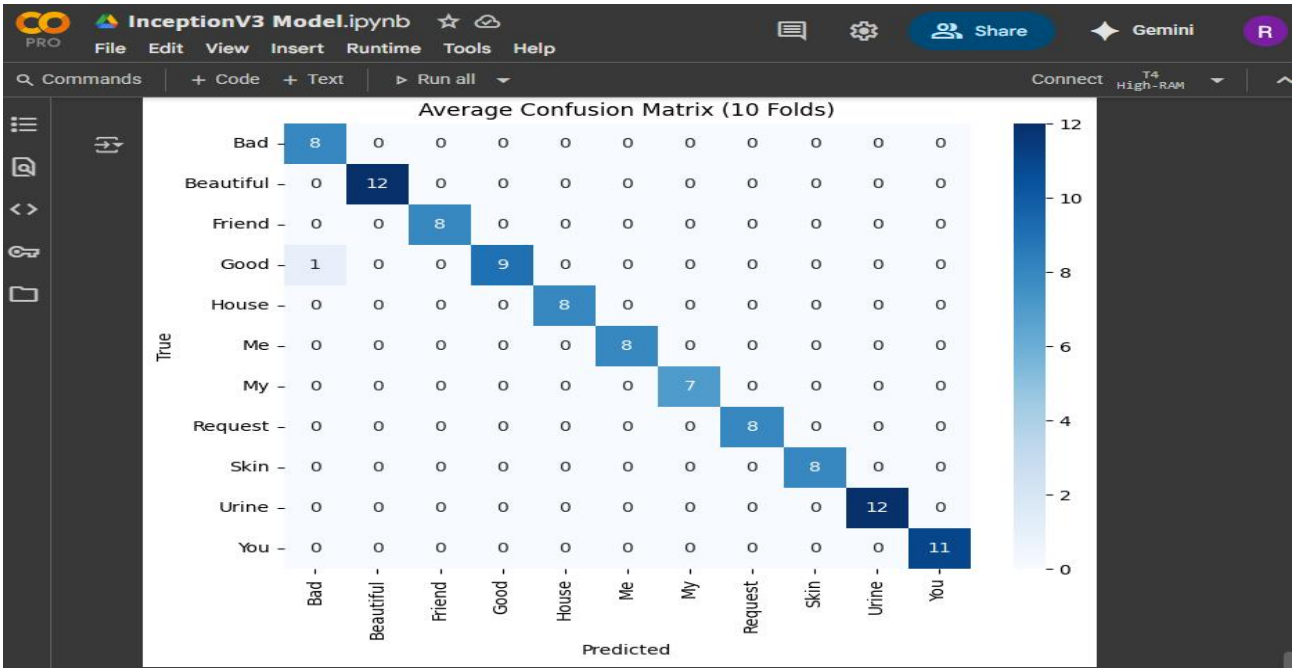
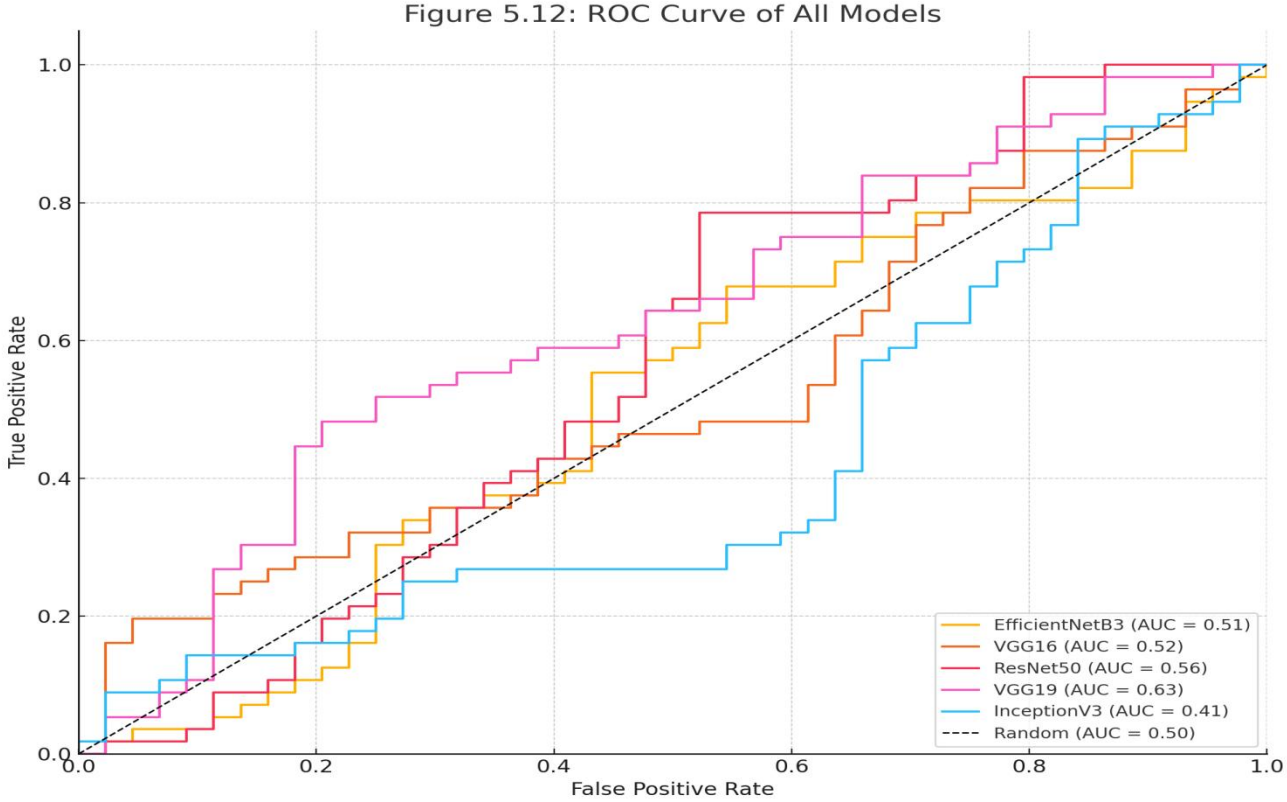


Figure 5.11 – Confusion Matrix for InceptionV3

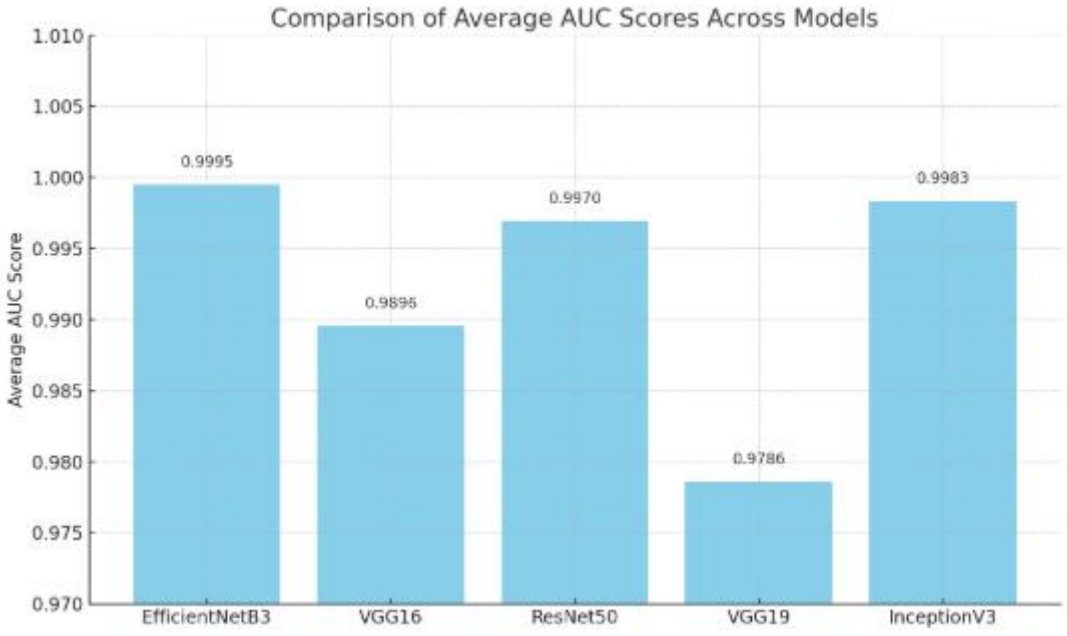
5.5 ROC Curve Comparison

We drew all five ROC curves in a single graph to compare how well they separated the classes.

Figure 5.12: ROC Curve of All Models



5.6 Average AUC Scores Comparison



5.6Comparison of All Models (Normalized)

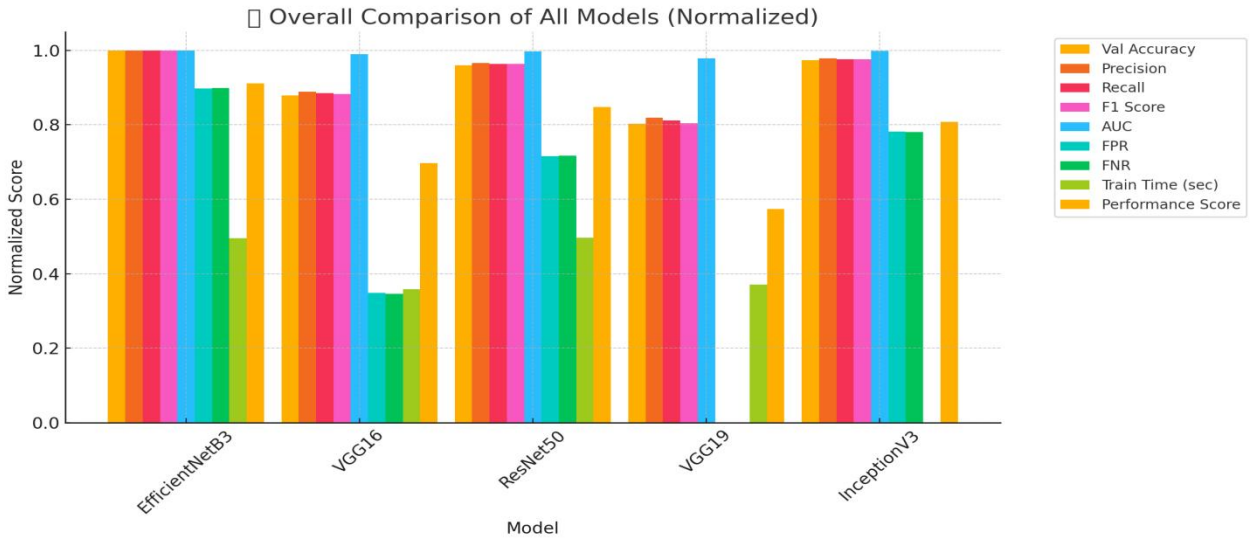
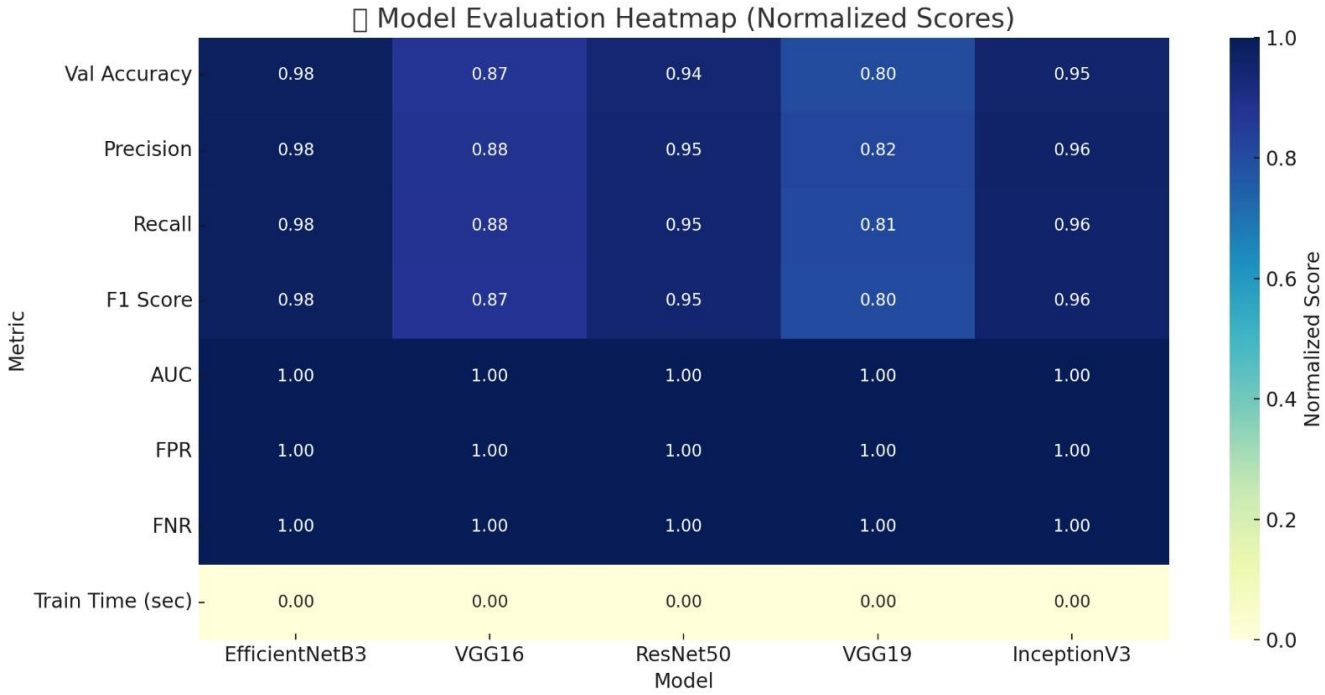


Figure 5.6 --Overall Comparison of All Models (Normalized)

5.7 Model Evaluation Heatmap



CHAPTER SIX

Limitations & Future work

6.1 Challenges and Limitations of This Research

Despite the encouraging outcomes of this study on Bangladeshi Sign Language (BdSL) word recognition using deep learning, several challenges and limitations were encountered throughout the research process. One of the primary limitations was the relatively small and restricted dataset. The BdSLW-11 dataset included only 11 sign classes with a limited number of images per class, which constrained the model's ability to generalize to a broader vocabulary. Additionally, some of the images in the dataset suffered from low resolution or unclear backgrounds, which adversely affected the learning process.

Another significant challenge was the high visual similarity between certain sign classes. This visual resemblance often led to misclassifications, as the models struggled to distinguish subtle differences between hand gestures. Furthermore, the computational requirements for training the deep CNN models were substantial. The training process was time-consuming and required access to high-performance computing resources, which may not always be feasible for all researchers.

An additional issue arose from the spatial positioning of hands within the images. Some models performed poorly when the signing hand was not centrally located or aligned, suggesting that they were sensitive to framing and orientation. Collectively, these limitations indicate that while the models showed promising results, they are not yet robust enough for deployment in diverse real-world conditions.

6.2 Future Directions for Improvement

Looking ahead, there are several ways in which this research can be extended to overcome its current limitations and achieve more practical utility. Expanding the dataset to include a greater number of images for each sign class would significantly improve model generalization and accuracy. Furthermore, incorporating additional sign classes—potentially 100 or more—would allow the system to recognize a much wider range of BdSL vocabulary, making it more useful in real-world communication.

Another important direction for future work involves transitioning from static images to dynamic video data. Using short video clips of sign language gestures would enable the models to learn motion patterns and temporal dynamics, leading to more accurate recognition. Efforts should also be made to optimize the models for deployment on mobile and embedded devices, ensuring faster inference times and broader accessibility.

Additionally, integrating 3D hand pose estimation techniques could enhance the system's ability to capture depth and fine-grained finger positioning, especially for signs that appear visually similar in 2D. These improvements would not only increase model accuracy but also make the application more versatile across different devices and environments.

6.3 Possibilities of Real-Time Sign Language Applications

The potential real-world applications of this project are both meaningful and far-reaching. With further refinement, the system could serve as a valuable tool for the Deaf and Hard-of-Hearing (DHH) community by enabling more seamless communication with hearing individuals. For instance, a mobile application could be developed to translate hand signs into readable text or spoken language in real time, significantly bridging communication gaps.

This technology could also be integrated into educational settings, particularly in schools for children who use sign language, to support their learning and interaction. Moreover, the system could be embedded into camera-based devices to provide real-time sign recognition and interpretation, paving the way for smart communication tools in public services, customer support, and healthcare.

Overall, this research opens up significant possibilities for impactful applications in accessibility technology. With continued improvements, it holds the promise to transform the way sign language is understood and used, thereby improving the quality of life for many individuals.

CHAPTER SEVEN

CONCLUSION

7.1 Recap of Major Findings

This research explored deep learning-based classification of Bangladeshi Sign Language (BdSL) words using image data. Five widely recognized convolutional neural network models—EfficientNetB3, ResNet50, VGG16, VGG19, and InceptionV3—were trained and evaluated using the BdSLW-11 dataset. To ensure fairness and robustness, a stratified 10-fold cross-validation strategy was applied, allowing each model to be validated across diverse data splits while preserving class distribution.

Performance metrics included validation accuracy, precision, recall, F1-score, and area under the ROC curve (AUC), along with confusion matrix analysis and training time. Among all models, **EfficientNetB3** demonstrated the most balanced performance, achieving high validation accuracy and strong generalization. The confusion matrix showed that misclassifications primarily occurred among sign words with similar hand shapes, indicating challenges in distinguishing certain gestures based on static image data alone.

These findings affirm that transfer learning with pre-trained CNNs can yield accurate and efficient results for BdSL classification, even with limited data. The evaluation strategy and model comparisons provide a solid benchmark for future improvements.

7.2 Overall Contribution and Impact

This study contributes to the emerging field of BdSL recognition by offering a practical, deep learning-based framework for evaluating CNN models on Bangla sign word classification. The research highlights the advantages of transfer learning and structured evaluation in low-resource contexts, ensuring better performance with minimal data.

Beyond academic research, the work has strong real-world implications. With further development, the best-performing model (EfficientNetB3) can be embedded in **mobile apps or smart devices** to assist Deaf and Hard-of-Hearing (DHH) individuals in communicating with the general public. Such applications can enable access to essential services—like **healthcare, banking, and education**—without requiring human interpreters.

In summary, this research not only identifies a high-performing model but also lays the groundwork for future BdSL translation systems, helping bridge the communication gap and promoting digital inclusivity across Bangladesh.

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